



YFG HOMOLOGATION TRUNK.

Operator / Maintenance Manual

Manufactured by:
Paveway Automation Pvt. Ltd.
Sanghvi Steel Complex,
128/2, TELCO Road,
Chinchwad,
Pune - 411019



YFG HOMOLOGATION TRUNK

Marelli Motherson Automotive Lighting India Pvt Ltd.
Gat No-165/1,165/2 &165/3 Village Ambethan,
Dwarka School, Taluka Khed, Chakan, Pune-410501



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SYSTEM INFORMATION:

Name :

YFG HOMOLOGATION TRUNK.

Customer :

Marelli Motherson Automotive Lighting India Pvt Ltd.
Gat No-165/1,165/2 &165/3 Village Ambethan,
Dwarka School, Taluka Khed, Chakan, Pune-410501.

Manufacturing Date :

June, 2022.

Supply Voltage :

230 VAC, 50/60 Hz

Power consumption:

1.105KW

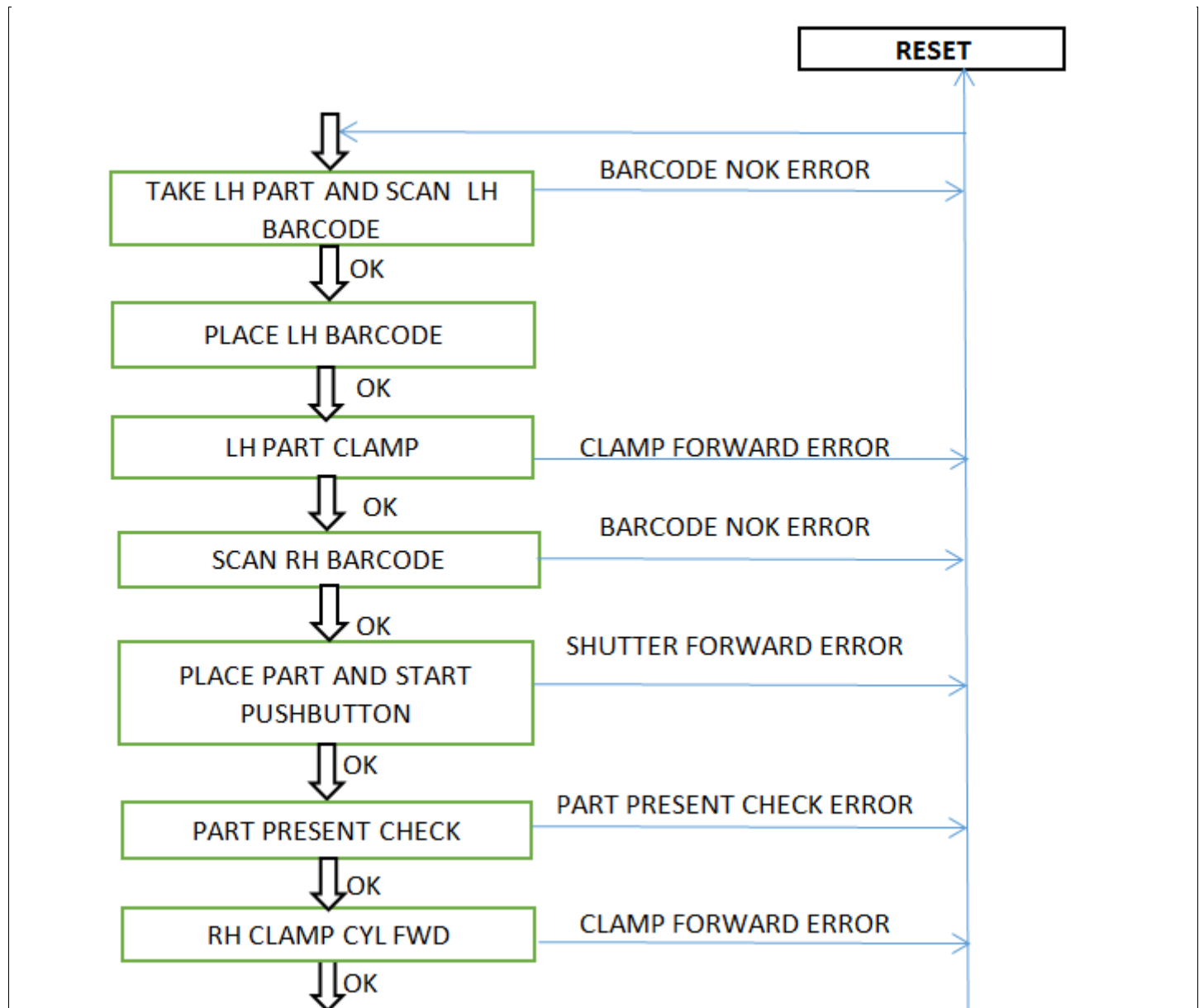
Manufacturer :

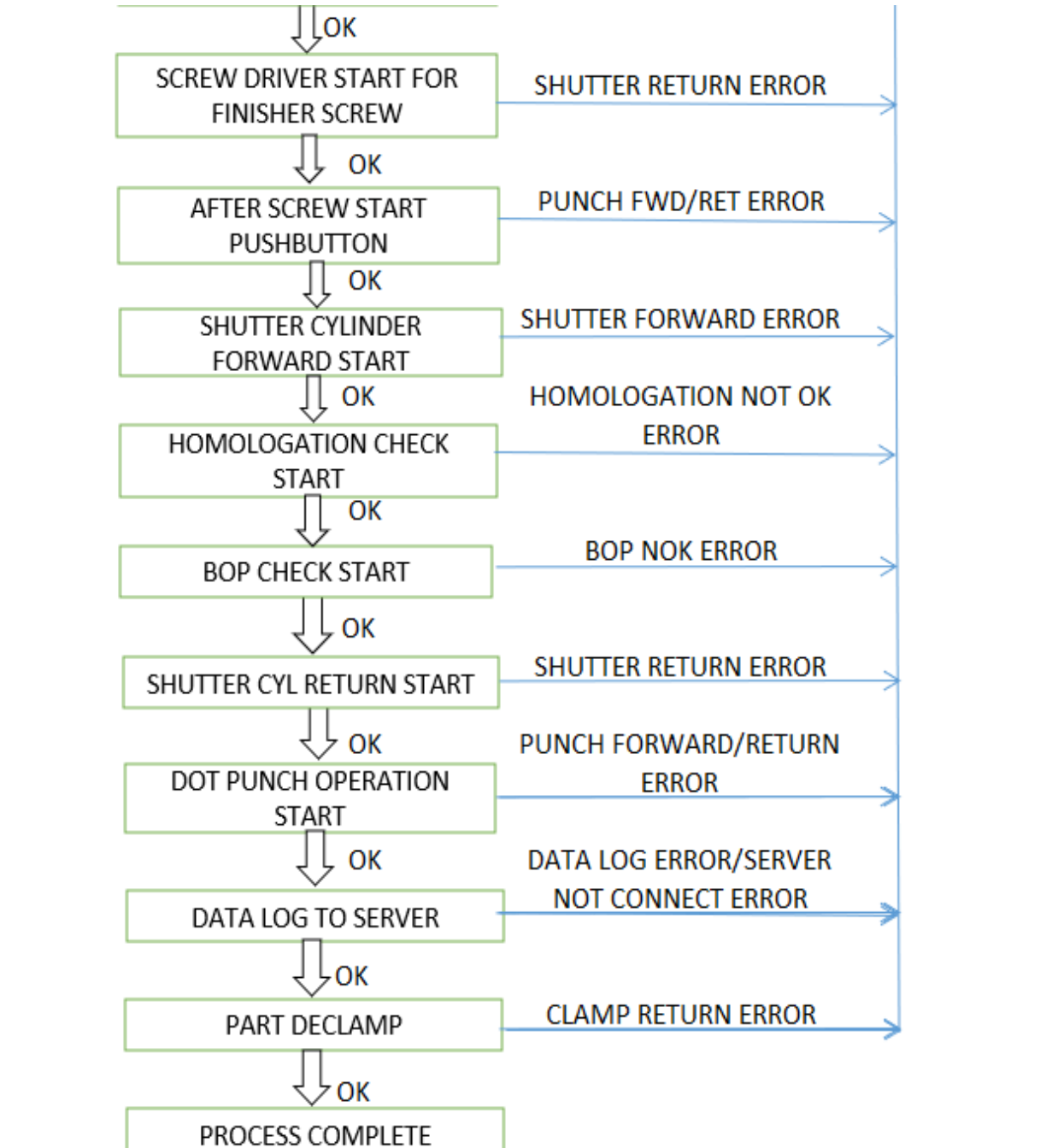
Paveway Automation Pvt. Ltd.
Sanghvi Steel Complex,
128/2, Telco Road, Chinchwad,
Pune- 411019.
tel: 020-27450840

Email ID :

applications@pavewayautomation.com

PROCESS SEQUENCE:

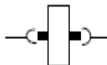
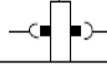
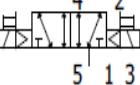
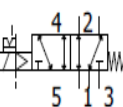




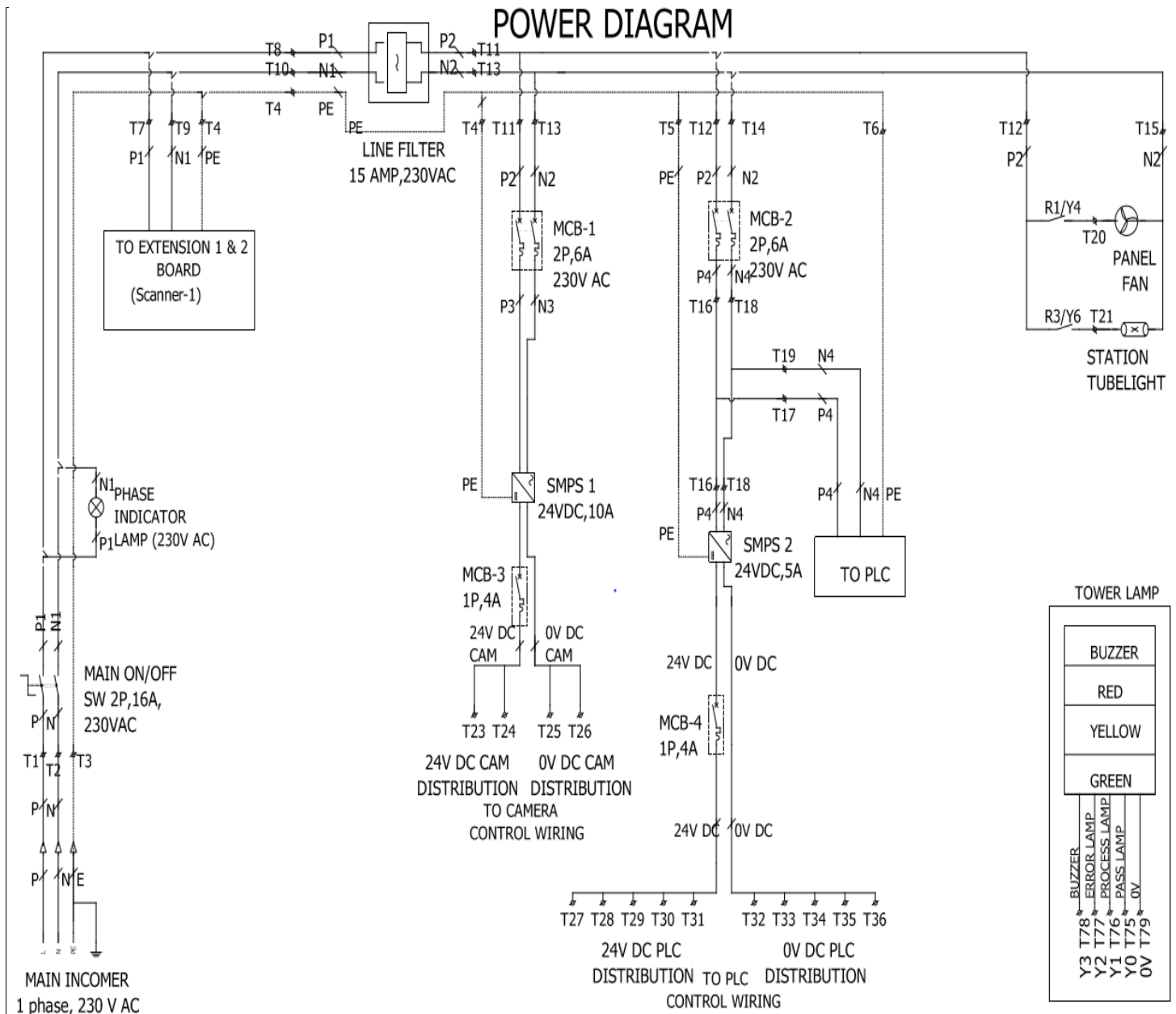
WIRING DIAGRAMS:

SYMBOL DESCRIPTION

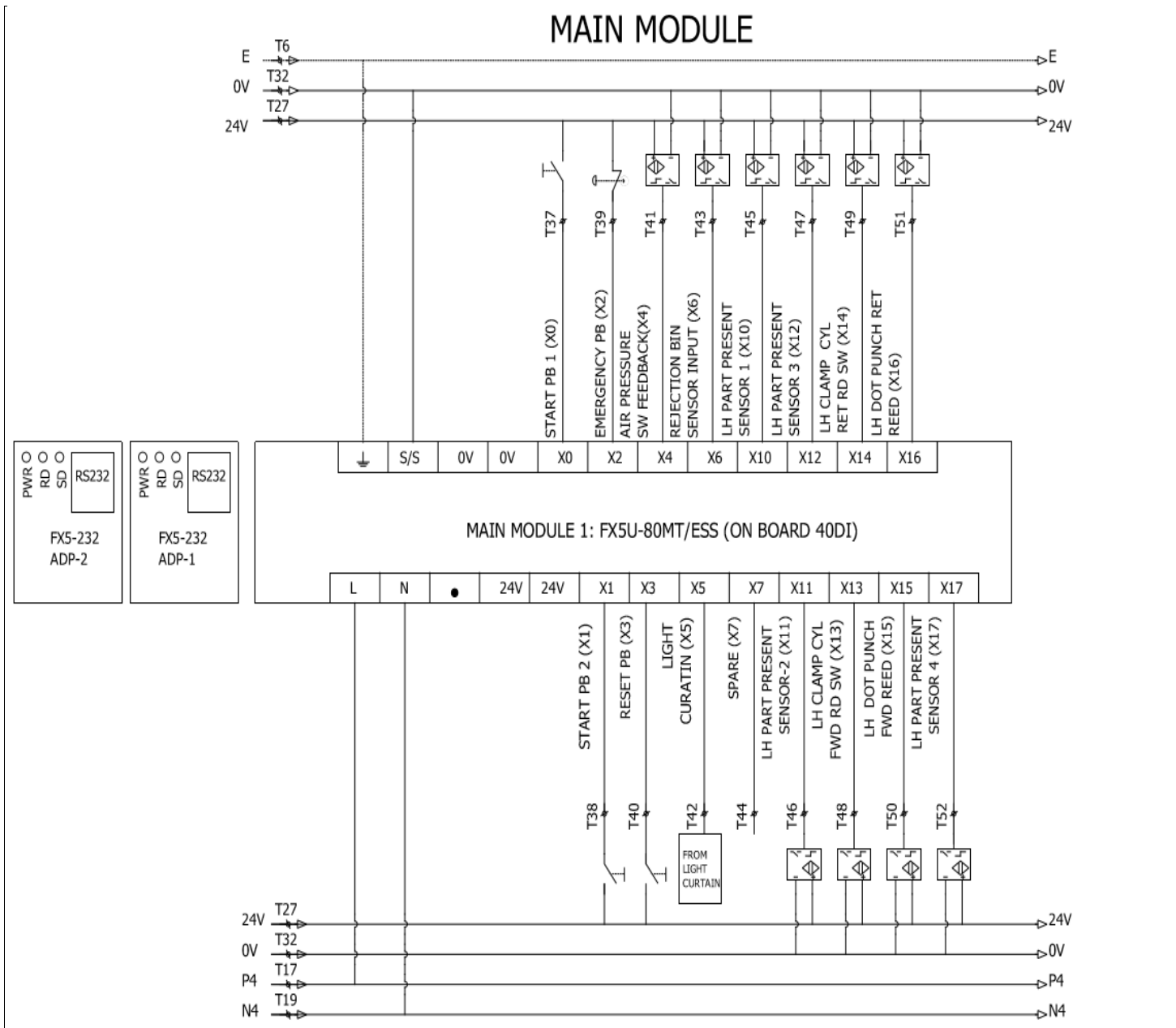
SYMBOL DESCRIPTION

Symbol	Symbol description	Symbol	Symbol description
	Earthing		Proximity sensor
	2 Pole MCB		Buzzer
	2 Pole on/off switch		Double acting solenoid valve
	Lamp		
	Tubelight		
	Fan		
	SMPS		Single acting solenoid valve (pneumatic)
	1 pole MCB		single acting solenoid valve
	Push button		
	Relay coil		
	Emergency stop button		
	Airoff		

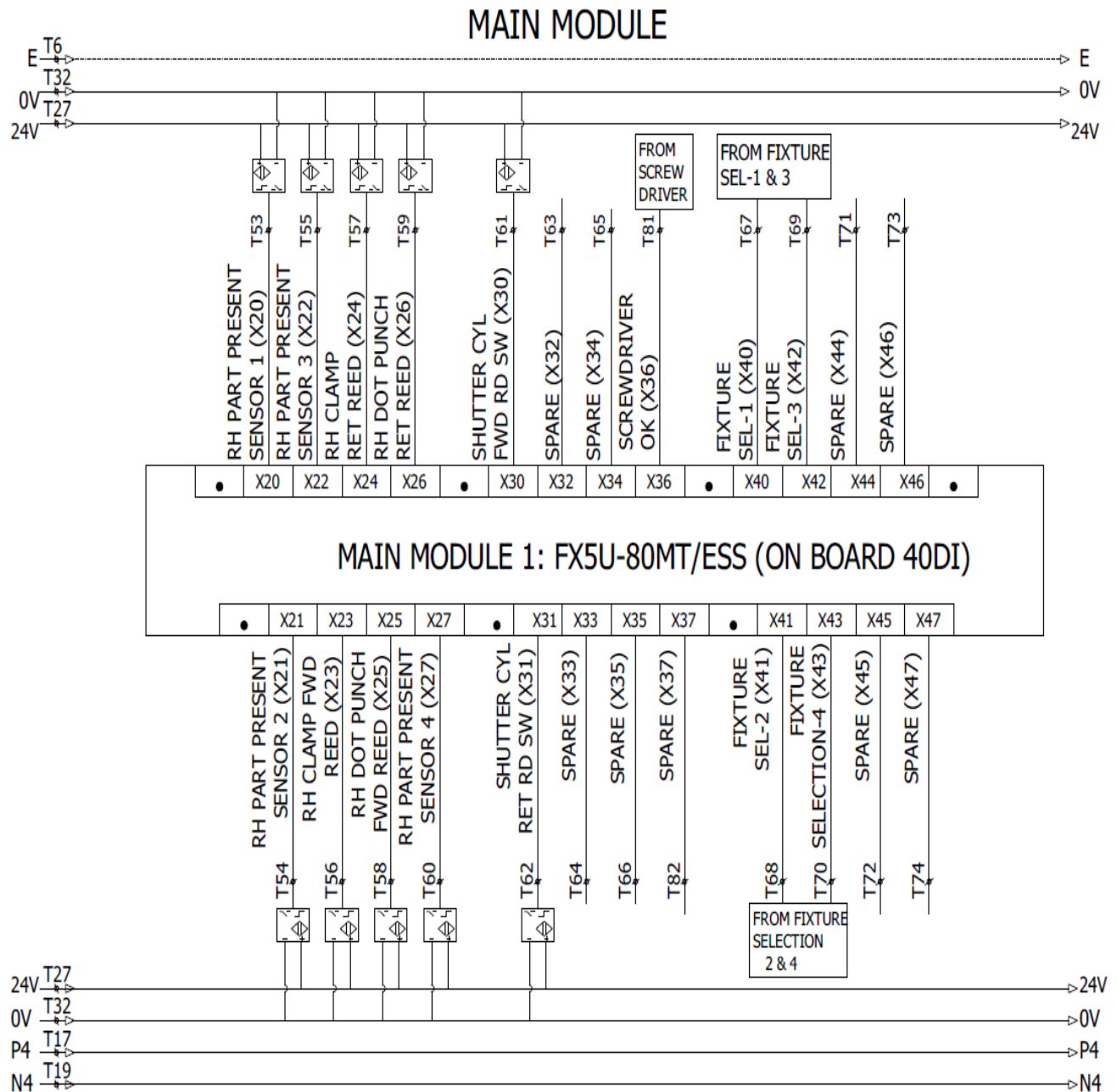
POWER DIAGRAM



MAIN MODULE 1:FX5U-80MT/ESS (ON BOARD 40DI)

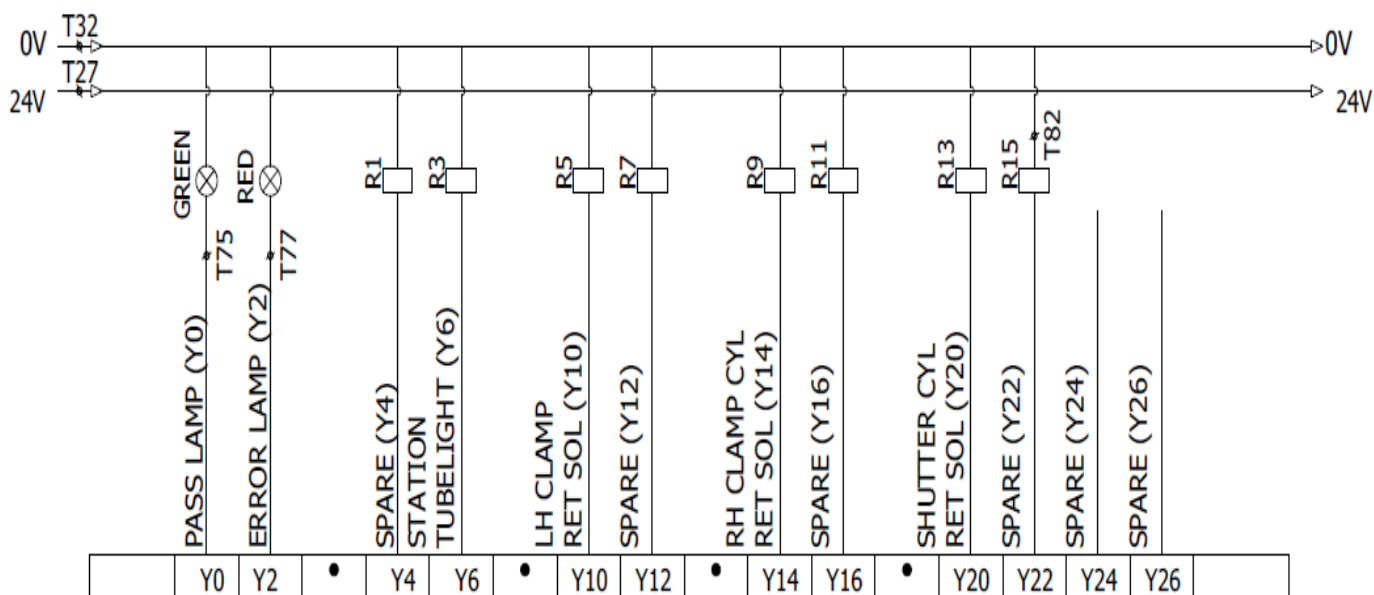


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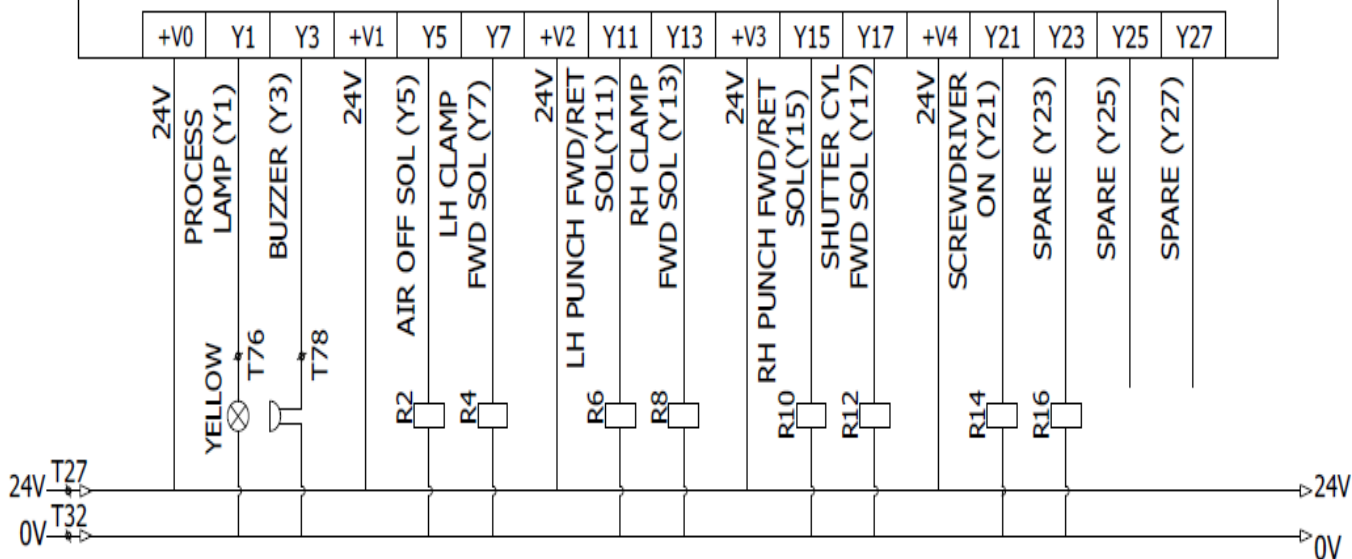


MAIN MODULE 1: FX5U-80MT/ESS (ON BOARD 40DO)

MAIN MODULE

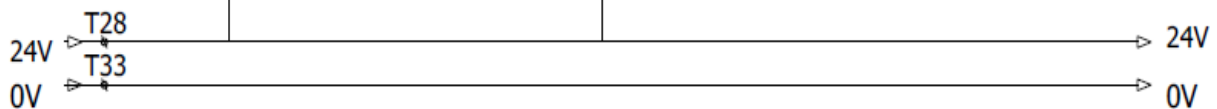
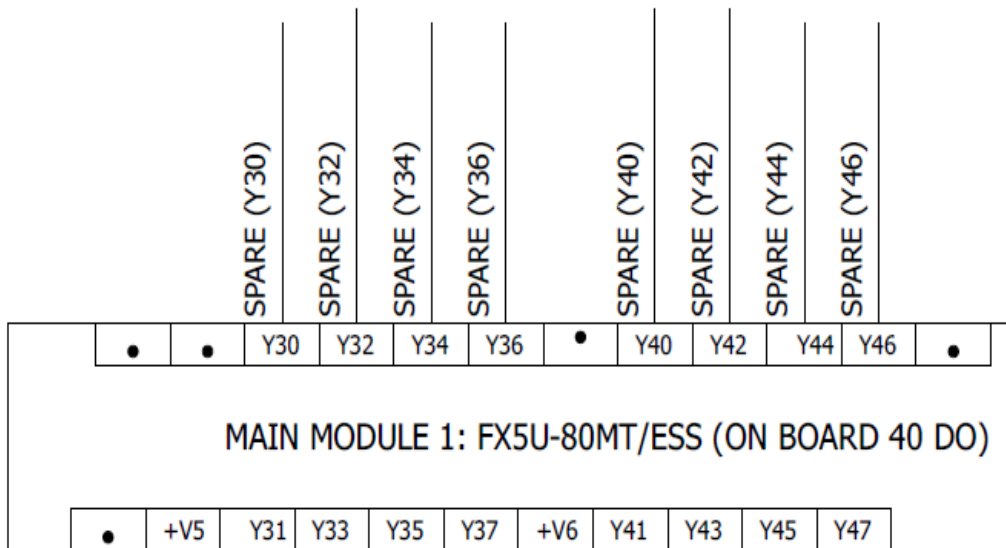
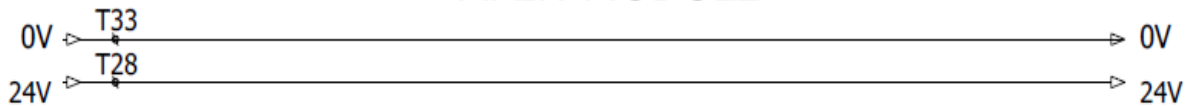


MAIN MODULE 1: FX5U-80MT/ESS (ON BOARD 40DO)



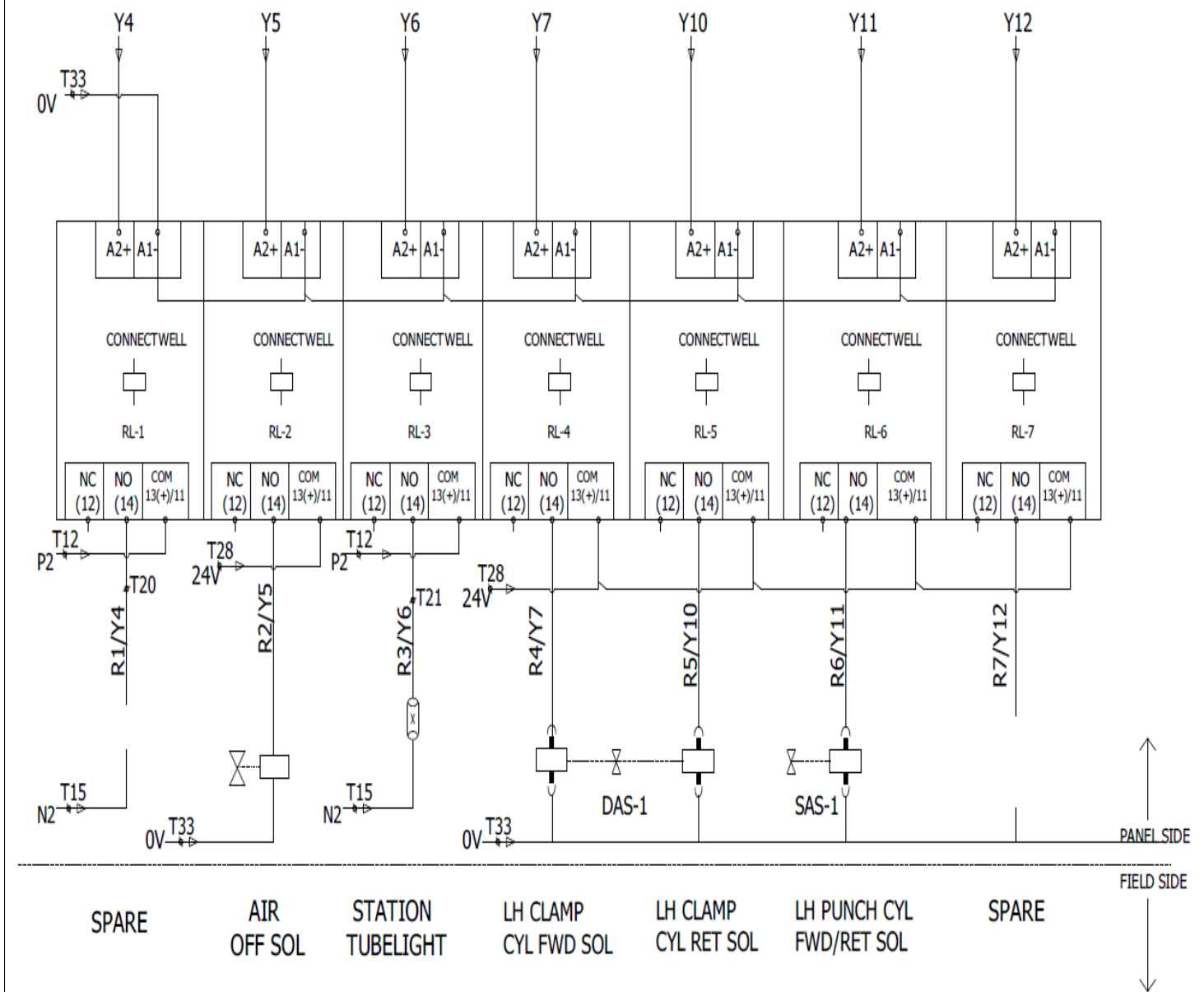
MAIN MODULE 1: FX5U-80MT/ESS (ON BOARD 40DO)

MAIN MODULE



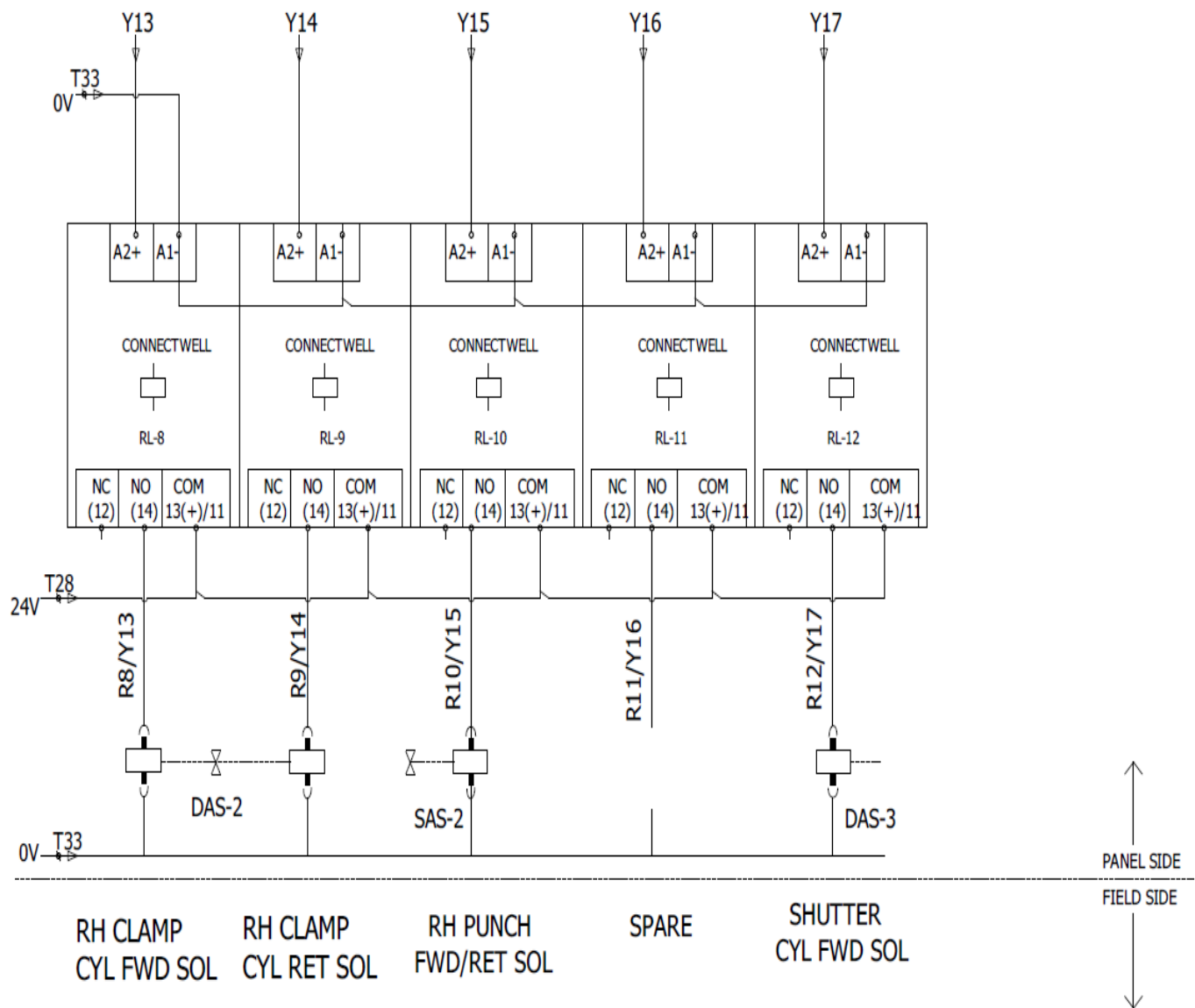
RELAY DIAGRAM-1

RELAY DIAGRAM-1



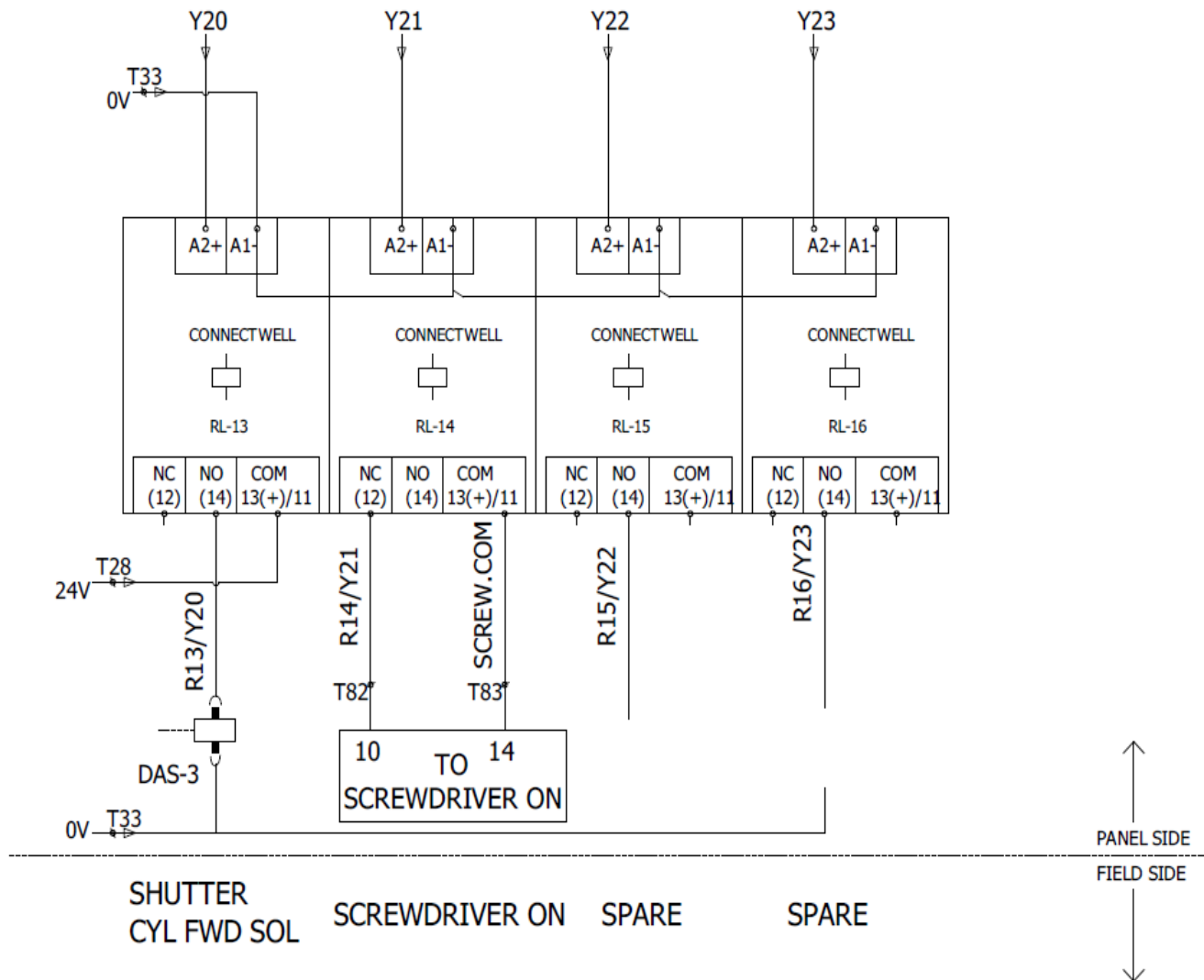
RELAY DIAGRAM-2

RELAY DIAGRAM-2



RELAY DIAGRAM-3

RELAY DIAGRAM-3



PANEL CONNECTOR DETAILS-1

PANEL CONNECTOR DETAILS-1

FIELD SIDE

ENDSTOPPER	1	P	PHASE DISTRIBUTION
	2	N	NEUTRAL DISTRIBUTION
ENDPLATE	3	E	EARTH DISTRIBUTION
	4	E	
	5	E	
	6	E	
ENDPLATE	7	P1	PHASE DISTRIBUTION
	8	P1	
	9	N1	NEUTRAL DISTRIBUTION
	10	N1	
	11	P2	PHASE DISTRIBUTION
	12	P2	
ENDPLATE	13	N2	NEUTRAL DISTRIBUTION
	14	N2	
	15	N2	
ENDPLATE	16	P4	PHASE DISTRIBUTION
	17	P4	
ENDPLATE	18	N4	NEUTRAL DISTRIBUTION
	19	N4	
ENDPLATE	20	R1/Y4	SPARE
ENDPLATE	21	R3/Y6	TO STATION TUBELIGHT
ENDPLATE	22	E	EARTH DISTRIBUTION TO CAMERA 1 & 2
ENDPLATE	23	24VDC	24V DC DISTRIBUTION TO CAMERA CONTR
	24	24VDC	
ENDPLATE	25	0VDC	0V DC DISTRIBUTION TO CAMERA CONTR
	26	0VDC	
ENDPLATE			

PANEL SIDE



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PANEL CONNECTOR DETAILS-2

PANEL CONNECTOR DETAILS-2

FIELD SIDE

TO CONTROL WIRING	27	24VDC
	28	24VDC
	29	24VDC
	30	24VDC
	31	24VDC
TO CONTROL WIRING	32	0V
	33	0V
	34	0V
	35	0V
	36	0V
TO PUSH BUTTONS	37	X0
	38	X1
	39	X2
	40	X3
	41	X4
	42	X5
	43	X6
	44	X7
TO AIR PRESSURE SW LIGHT CURTAIN REJECTION BIN SENSOR SPARE	45	X10
	46	X11
	47	X12
	48	X13
	49	X14
	50	X15
	51	X16
	52	X17
FROM 24 PIN INDO CONNECTOR-LH	53	X20
	54	X21
	55	X22
	56	X23
	57	X24
	58	X25
	59	X26
	60	X27
FROM SMPS-2 24V DC, 5A	27	24VDC
	28	24VDC
	29	24VDC
	30	24VDC
	31	24VDC
FROM SMPS 24V DC, 5A	32	0V
	33	0V
	34	0V
	35	0V
	36	0V
START PB 1 START PB 2 EMERGENCY STOP RESET PB AIR PRESSURE SW LIGHT CURTAIN REJECTION BIN SENSOR SPARE	37	X0
	38	X1
	39	X2
	40	X3
	41	X4
	42	X5
	43	X6
	44	X7
LH PART PRESENT SENSOR-1 LH PART PRESENT SENSOR-2 LH PART PRESENT SENSOR-3 LH CLAMP CYL FWD RD SW LH CLAMP CYL RET RD SW LH DOT PUNCH FWD RD SW LH DOT PUNCH RET RD SW LH PART PRESENT SENSOR 4	45	X10
	46	X11
	47	X12
	48	X13
	49	X14
	50	X15
	51	X16
	52	X17
RH PART PRESENT SENSOR-1 RH PART PRESENT SENSOR-2 RH PART PRESENT SENSOR-3 RH CLAMP CYL FWD RD SW RH CLAMP CYL RET RD SW RH DOT PUNCH FWD RD SW RH DOT PUNCH RET RD SW RH PART PRESENT SENSOR 4	53	X20
	54	X21
	55	X22
	56	X23
	57	X24
	58	X25
	59	X26
	60	X27

PANEL SIDE

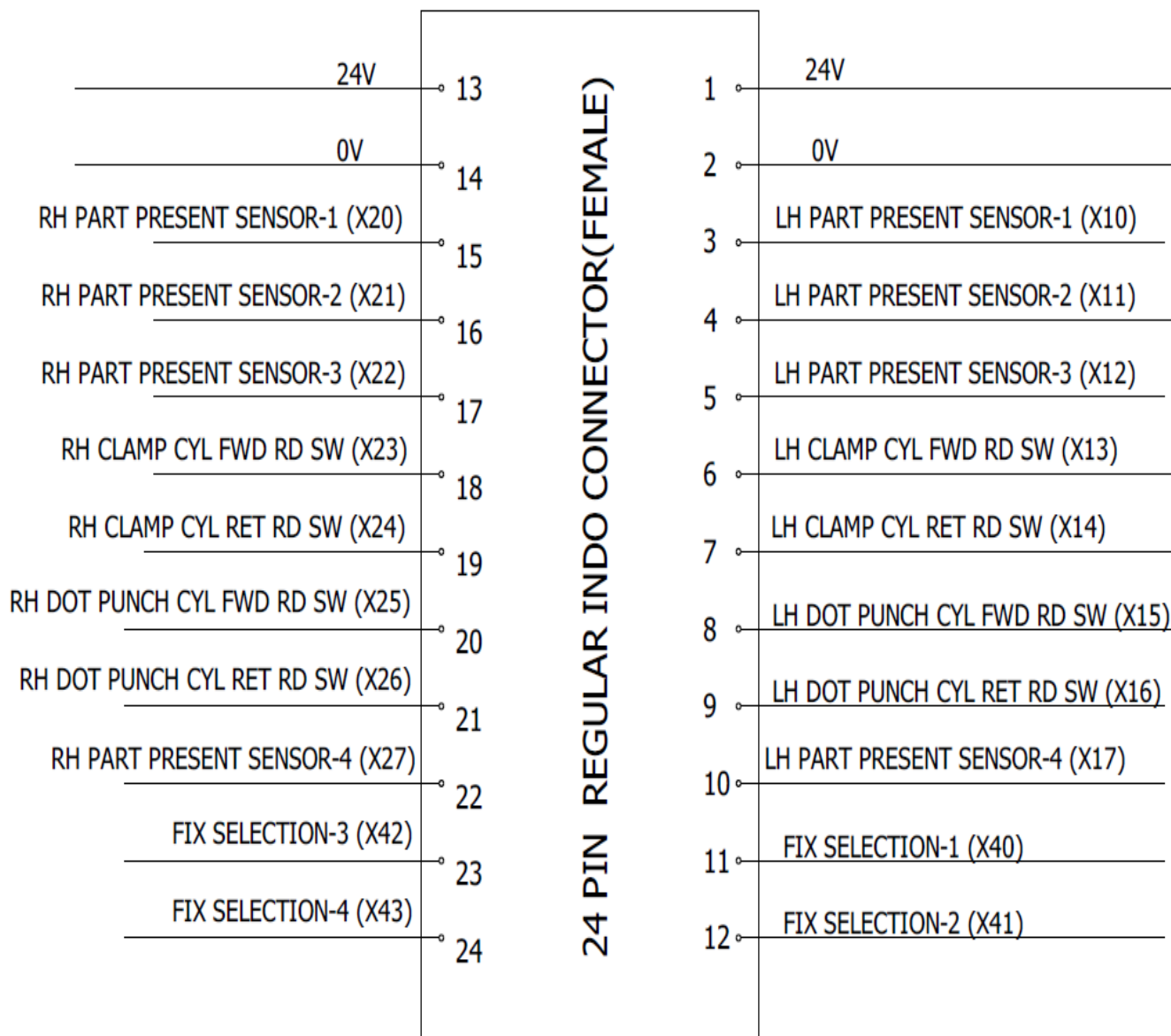
PANEL CONNECTOR DETAILS-3

PANEL CONNECTOR DETAILS-3

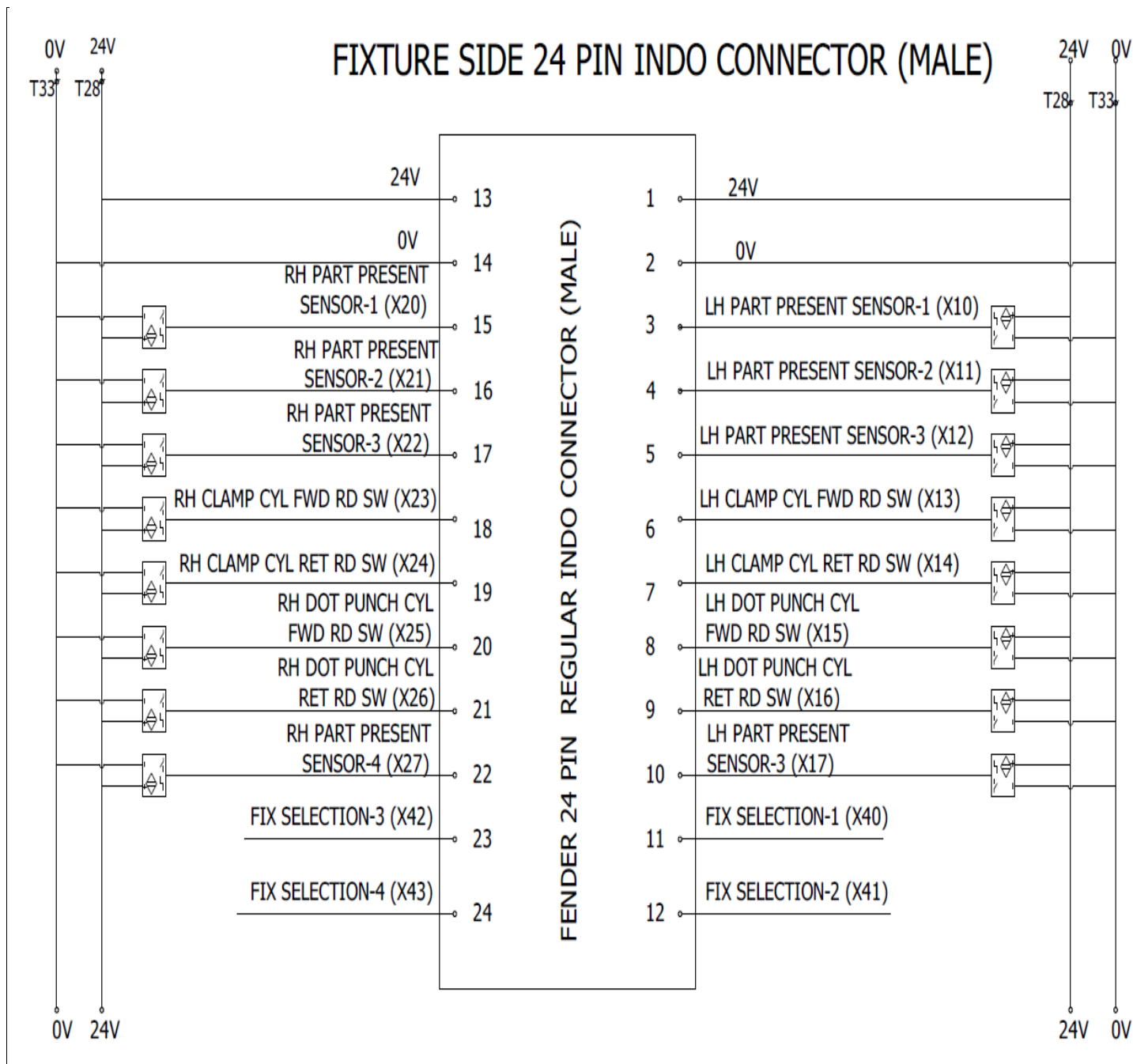
FIELD SIDE

ENDPLATE	61	X30	SHUTTER CYL FWD REED SW
	62	X31	SHUTTER CYL RET REED SW
	63	X32	SPARE
	64	X33	SPARE
	65	X34	SPARE
	66	X35	SPARE
ENDPLATE	67	X40	FIXTURE SELECTION-1
	68	X41	FIXTURE SELECTION-2
	69	X42	FIXTURE SELECTION-3
	70	X43	FIXTURE SELECTION-4
	71	X44	SPARE
	72	X45	SPARE
ENDPLATE	73	X46	SPARE
	74	X47	SPARE
	75	Y0	PASS LAMP
	76	Y1	PROCESS LAMP
	77	Y2	ERROR LAMP
	78	Y3	BUZZER
ENDPLATE	79	0V	0V
	80	X36	SCREW DRIVER OK
	81	X37	SPARE
	82	R15/Y22	SCREWDRIVER ON
	83	SCREW. COM	SCREWDRIVER COM
	84	24V	SCREWDRIVER 24V
ENDSTOPPER			

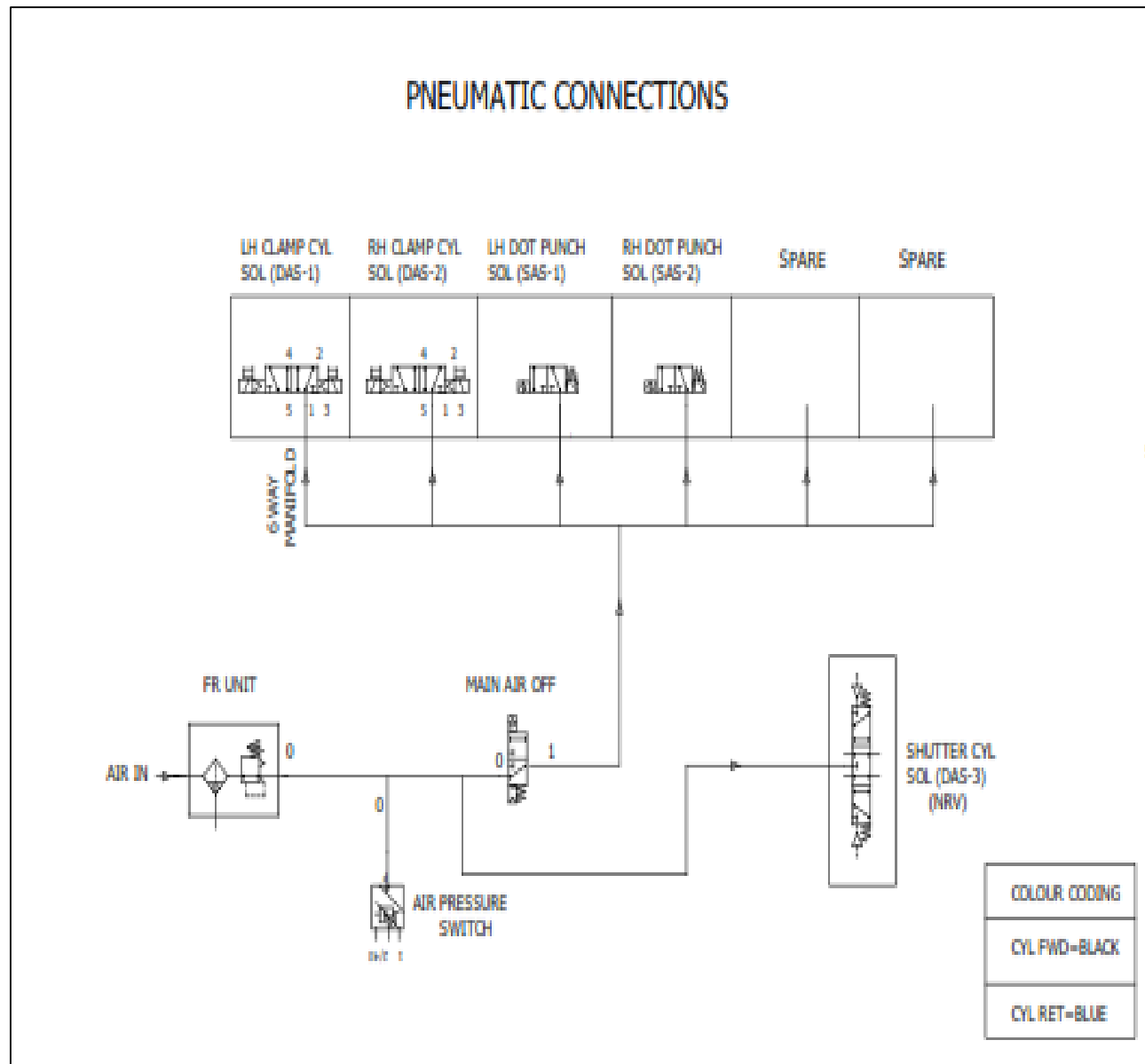
PANEL SIDE

MACHINE SIDE 24 PIN INDO CONNECTOR (FEMALE)
MACHINE SIDE 24 PIN INDO CONNECTOR(FEMALE)


FIXTURE SIDE 24 PIN INDO CONNECTOR (MALE)



PNEUMATIC CONNECTIONS





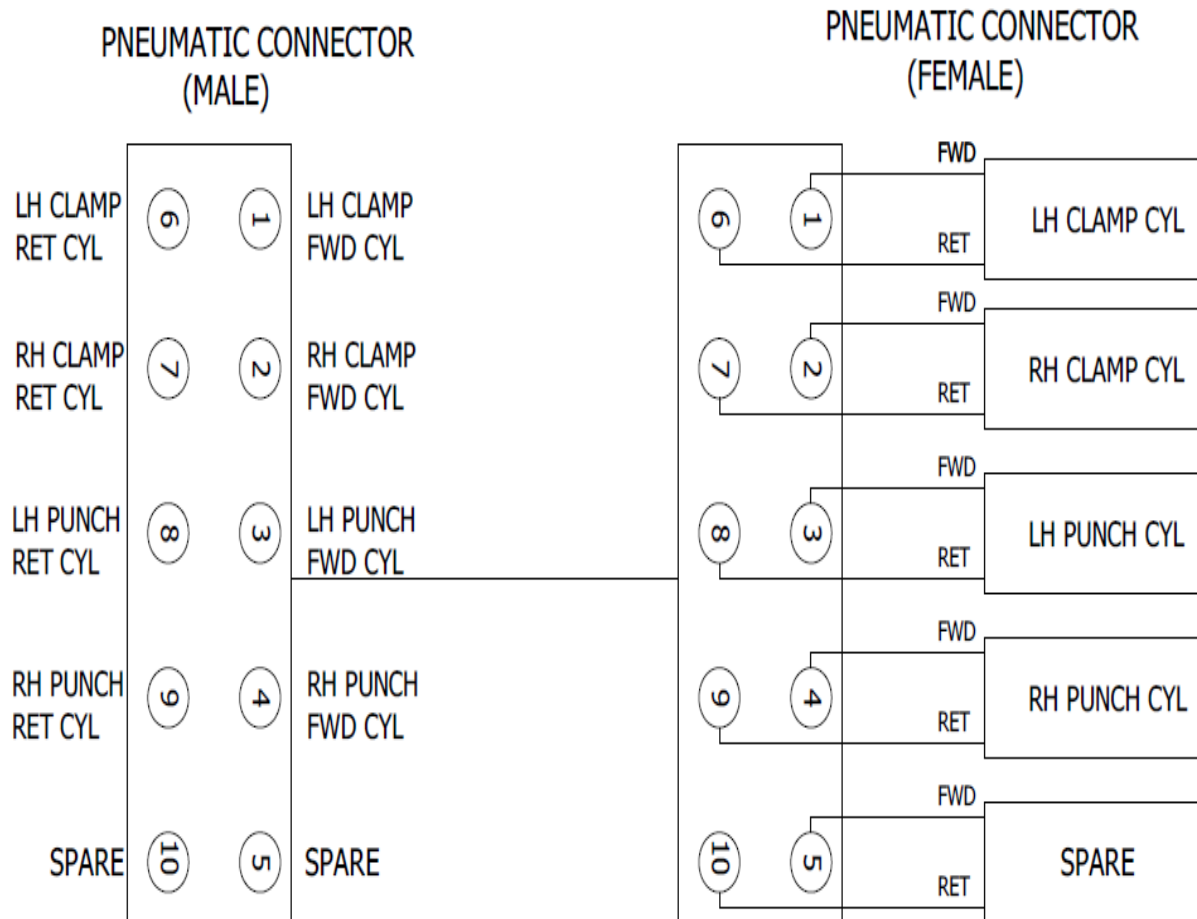
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PNEUMATIC LAYOUT

PNEUMATIC LAYOUT





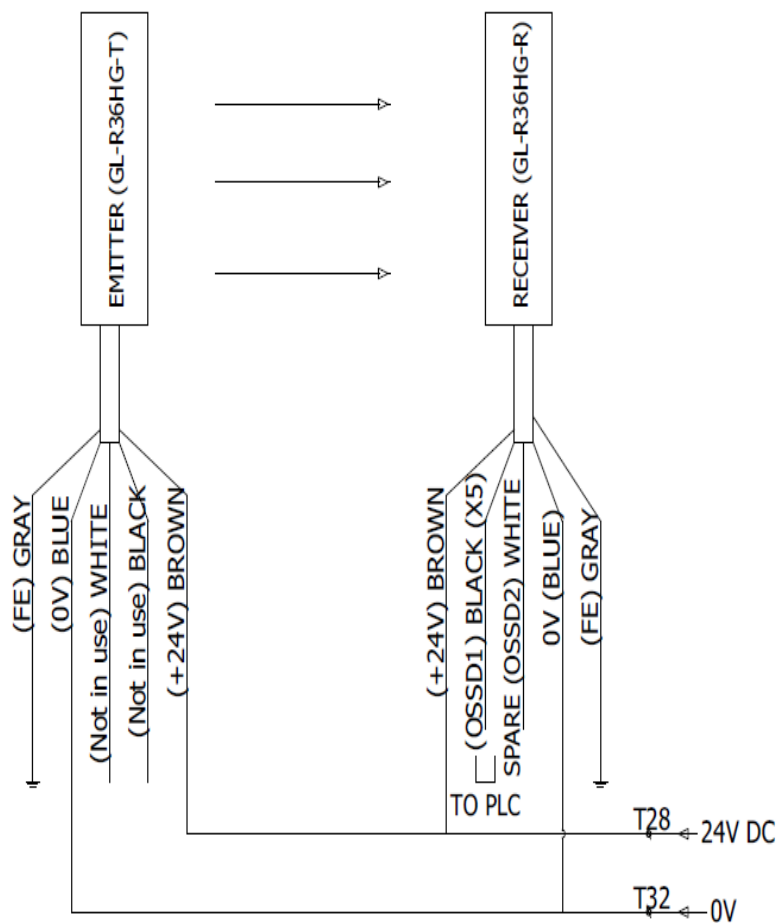
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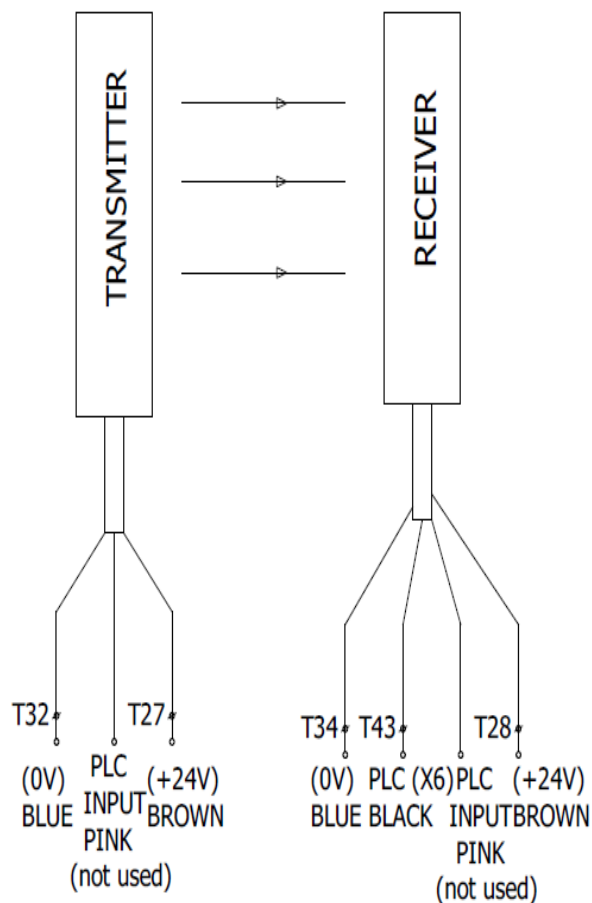


LIGHT CURTAIN CONNECTION& REJECTION BIN CONNECTION

LIGHT CURTAIN CONNECTIONS (KEYENCE:-GL-R36HG)



REJECTION BIN SENSOR CONNECTIONS (PANASONIC:-NA1-5-PN)





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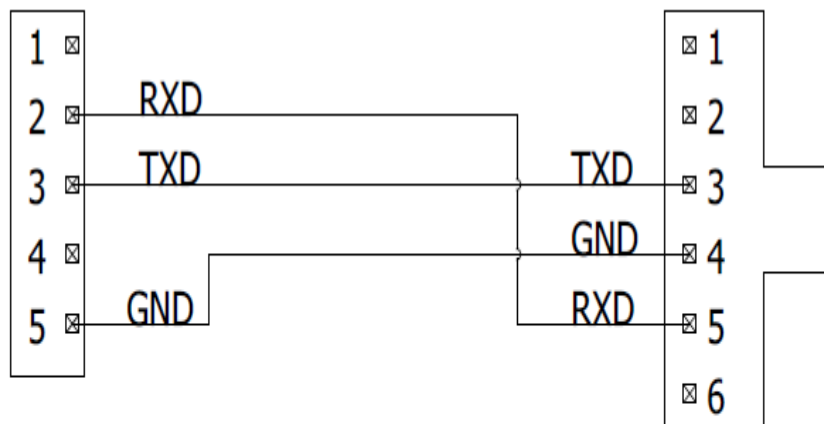


CAMERA CONTROLLER CABLE DETAILS (CV-X320F)

CAMERA CONTROLLER CABLE DETAILS(CV-X320F)

mitsubishi FX5-232ADP
(DB9 CONNECTOR)

CV-X320F
CONTROLLER
(RJ11)





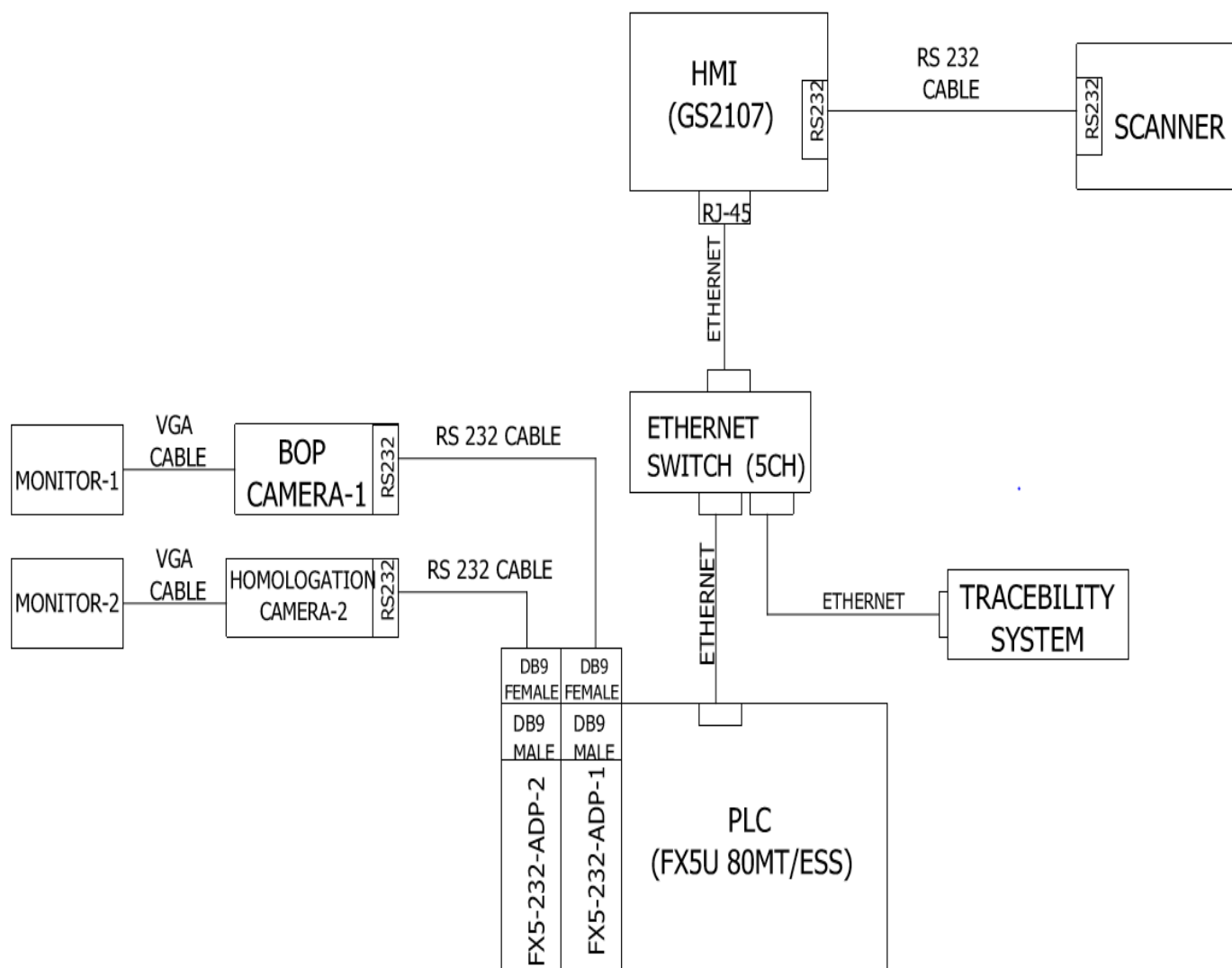
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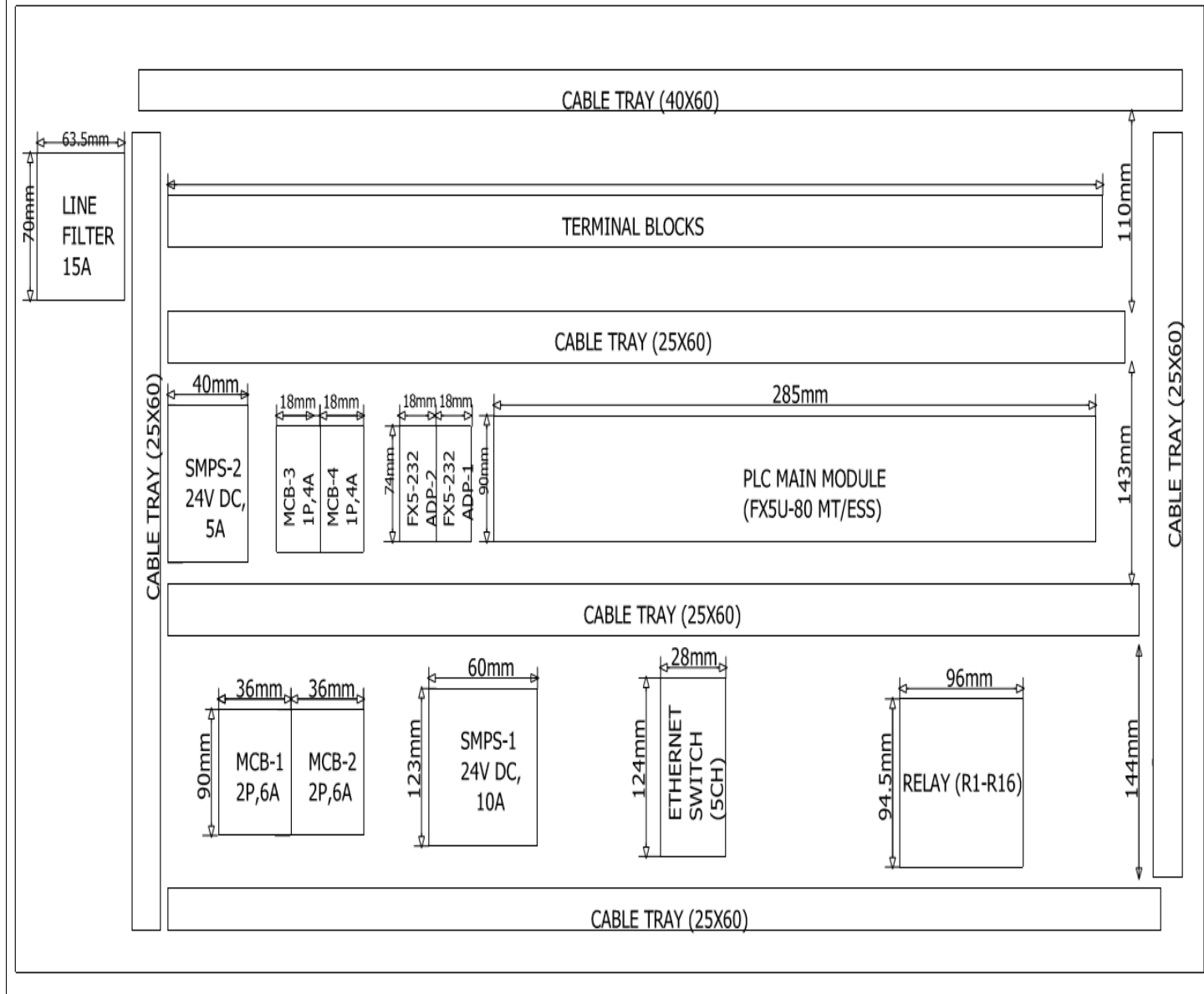
COMMUNICATION ARCHITECTURE

COMMUNICATION ARCHITECTURE



COMPONENT LAYOUT

COMPONENT LAYOUT





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BILL OF MATERIAL:

BOM

ELECTRICAL PANEL MATERIAL

SN	COMPONENT NAME	DESCRIPTION	PART NO.	MAKE	QTY
1	Control panel	600X600X300	MAS0606030R5	Eldon	1
2	Plc	40DI/40DO	FX5U-80MT/ESS	Mitsubishi	1
3	HMI	7"	GS2107	Mitsubhishi	1
4	RS232 module	Communication module	FX5U-232ADP	Mitsubishi	2
5	Smps	24 VDC,10A	S8VK-C24024	Omron	1
6	Smps	24 VDC,5A	SPB-120-24	Autonics	1
7	On/off switch	2P,16A,230VAc	61002SAB13TDYR	L&t	1
8	Mcb	2P,6A	5SL42167RC	siemens	2
9	Mcb (24vdc)	1P,4A	5SL61047RC	siemens	2
10	Push button (green)	Green Mashrum Type	2AM3	Teknic	2
11	Reset Button(red)	Red Flush Type	2AM4	Teknic	1
12	Emergency stop	Latched	2AML4	Teknic	1
13	No element	24 VDC	S1	Teknic	3
14	Nc element	24 VDC	S2	Teknic	1
15	Pb box	2Way	50MM	Teknic	2
16	Indication lamp (green)	230 VAC	3PLBR-3L	Teknic	1
17	Relay	24V,6A	SOCY1-24D	Connectwell	16
18	Tower lamp 3-tier	Steady,24 VDC		Patellite	1
19	Light curtain		GL-R36-HG	Keyence	1
20	Electric indo	24 Pin Regular, male female with housing & HOOD	for LH & RH	Indo	2
21	Panel fan	230VAC,18W,0.10A,2650RPM	12A230HSAC	Hicool	1
22	Led tube light (station)	230VAC	LP-201-15W	Lord's	2
23	Extension board & switch	6A(Scanner-1,PC)		Legrand	2
24	Ethernet hub	5-PORT,10-30VDC,70MA		Sixnet	1
25	Line filter	230VAC,15A	EF-1B015B01E-C02	Elcom	1
26	Industrial socket	230VAC,16A	TYP 215	Kosmos	1
27	Scanner		Powerscan PD9531	Datalogic	1
28	Camera controller		CV-X320F	Keyence	2
29	Camera			Keyence	4
30	Camera cable			Keyence	4



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31	Monitor + vga cable				2
32	Rejection bin		NA1-5-PN	Panasonic	1
33	Sensors			Baumer	10

PNEUMATIC MATERIAL

Sr No	Component Name	Description	Part No.	Make	Qty.
1	Solenoid valve	Double Acting	SY7227-5LZ-02	SMC	2
		Single Acting	SY7120-5LZ-02	SMC	2
2	Manifold	6 Way manifold	SS5Y7-20-06	SMC	1
		4 Way manifold	SS5Y7-20-07	SMC	0
3	Read switch	24VDC, PNP		SMC	9
4	Air off solenoid valve	0.7MPa	VT307-5DZ1-02	SMC	1
5	Digital pressure switch	Pressure Display	ISE30AF-01-P-LA2	SMC	1
6	Frl unit	1/4"	AW20-02BG1-B	SMC	1
7	Quick coupling connector 10 pin	Male+female,10 pin	KDM10S-06	SMC	2
8	Nrv valve	5/3" VALVE	SY7320-5LZ-02	SMC	1
9	Manifold plug	1/4"		SMC	1
10	Pneumatic tube	Blue 6 OD		SMC	
		Black 6 OD		SMC	



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I/O LIST:

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MODULES	I/P ADDR	INPUTS	O/P ADDR	OUTPUTS	RELAY
FX5U-80MT/ESS	X0	Start pb 1	Y0	Pass lamp	
	X1	Start pb 2	Y1	Process lamp	
	X2	Emergency stop pb	Y2	Error lamp	
	X3	Reset pb	Y3	Buzzer	
	X4	Air pressure switch	Y4	Spare	R1/Y4
	X5	Light curtain	Y5	Air off sol	R2/Y5
	X6	Rejection bin sensor	Y6	Station light	R3/Y6
	X7	Spare	Y7	Lh clamp fwd sol	R4/Y7
	X10	Lh part present sensor 1	Y10	Lh clamp ret sol	R5/Y10
	X11	Lh part present sensor 2	Y11	Lh punch fwd/ret sol	R6/Y11
	X12	Lh part present sensor 3	Y12	Spare	R7/Y12
	X13	Lh clamp fwd reed	Y13	Rh clamp fwd sol	R8/Y13
	X14	Lh clamp ret reed	Y14	Rh clamp ret sol	R9/Y14
	X15	Lh dot punch fwd reed	Y15	Rh punch fwd/ret sol	R10/Y15
	X16	Lh dot punch ret reed	Y16	Spare	R11/Y16
	X17	Lh part present sensor 4	Y17	Shutter cyl fwd sol	R12/Y17
	X20	Rh part present sensor 1	Y20	Shutter cyl ret sol	R13/Y20
	X21	Rh part present sensor 2	Y21	Screwdriver on	R14/Y21
	X22	Rh part present sensor 3	Y22	Spare	R15/Y22
	X23	Rh clamp fwd reed	Y23	Spare	R16/Y23
	X24	Rh clamp ret reed	Y24	Spare	
	X25	Rh dot punch fwd reed	Y25	Spare	
	X26	Rh dot punch ret reed	Y26	Spare	
	X27	Rh part present sensor 4	Y27	Spare	
	X30	Shutter cyl fwd reed	Y30	Spare	
	X31	Shutter cyl ret reed	Y31	Spare	
	X32	Spare	Y32	Spare	
	X33	Spare	Y33	Spare	
	X34	Spare	Y34	Spare	
	X35	Spare	Y35	Spare	
	X36	Screwdriver ok	Y36	Spare	
	X37	Spare	Y37	Spare	
	X40	Fixture sel 1	Y40	Spare	
	X41	Fixture sel 2	Y41	Spare	



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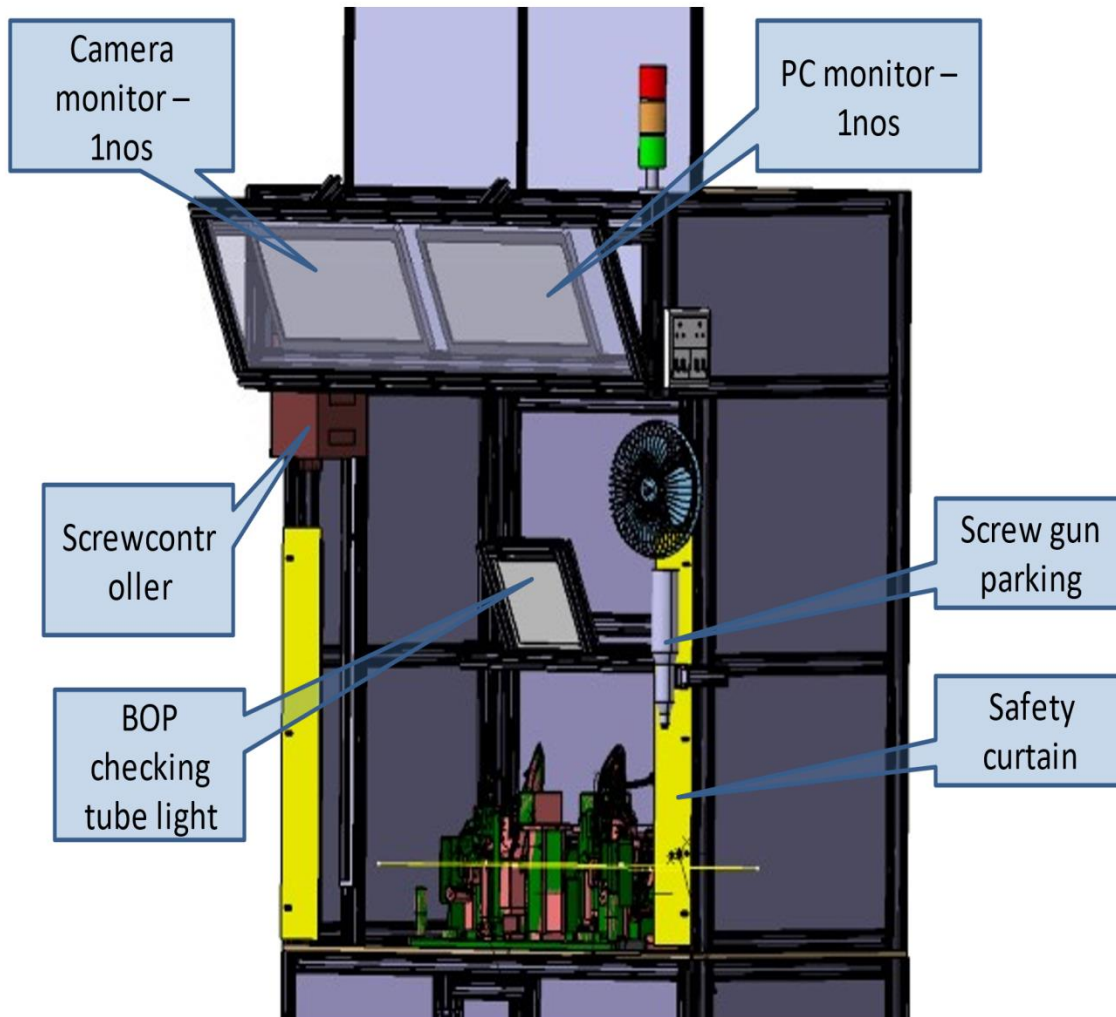
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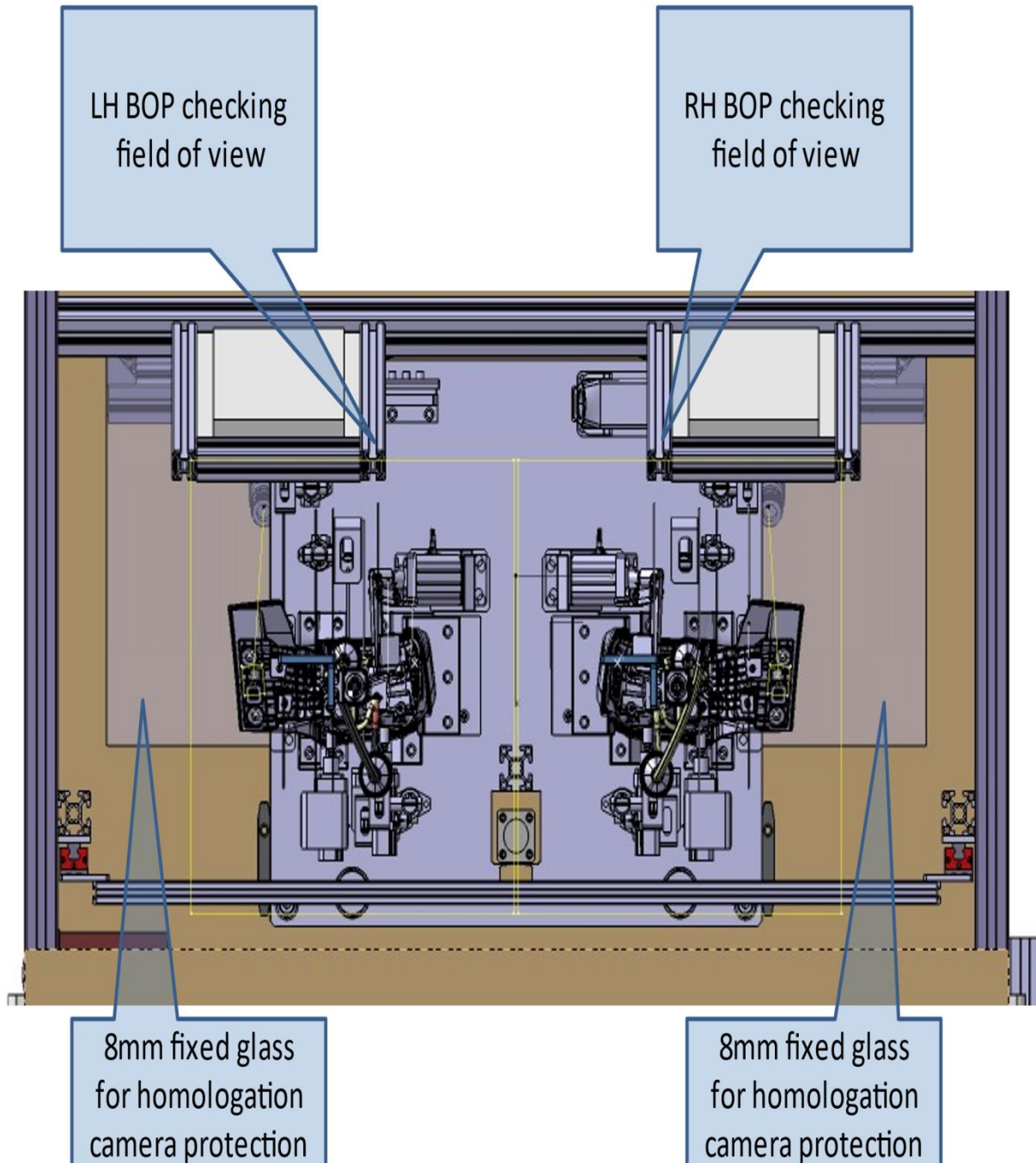
	X42	Fixture sel 3	Y42	Spare	
	X43	Fixture sel 4	Y43	Spare	
	X44	Spare	Y44	Spare	
	X45	Spare	Y45	Spare	
	X46	Spare	Y46	Spare	
	X47	Spare	Y47	Spare	
COMMUNICATION	PLC - HMI	ETHERNET			
	HMI - SCANNER	HMI RS232 PORT			
	PLC- CAMERA 1	FX5-232-ADP			
	PLC- CAMERA 2	FX5-232-ADP			
	PLC TRACEABILITY	ETHERNET			

MECHANICAL DETAILS:

COMPONENT DESCRIPTION



FIXTURE DETAILS



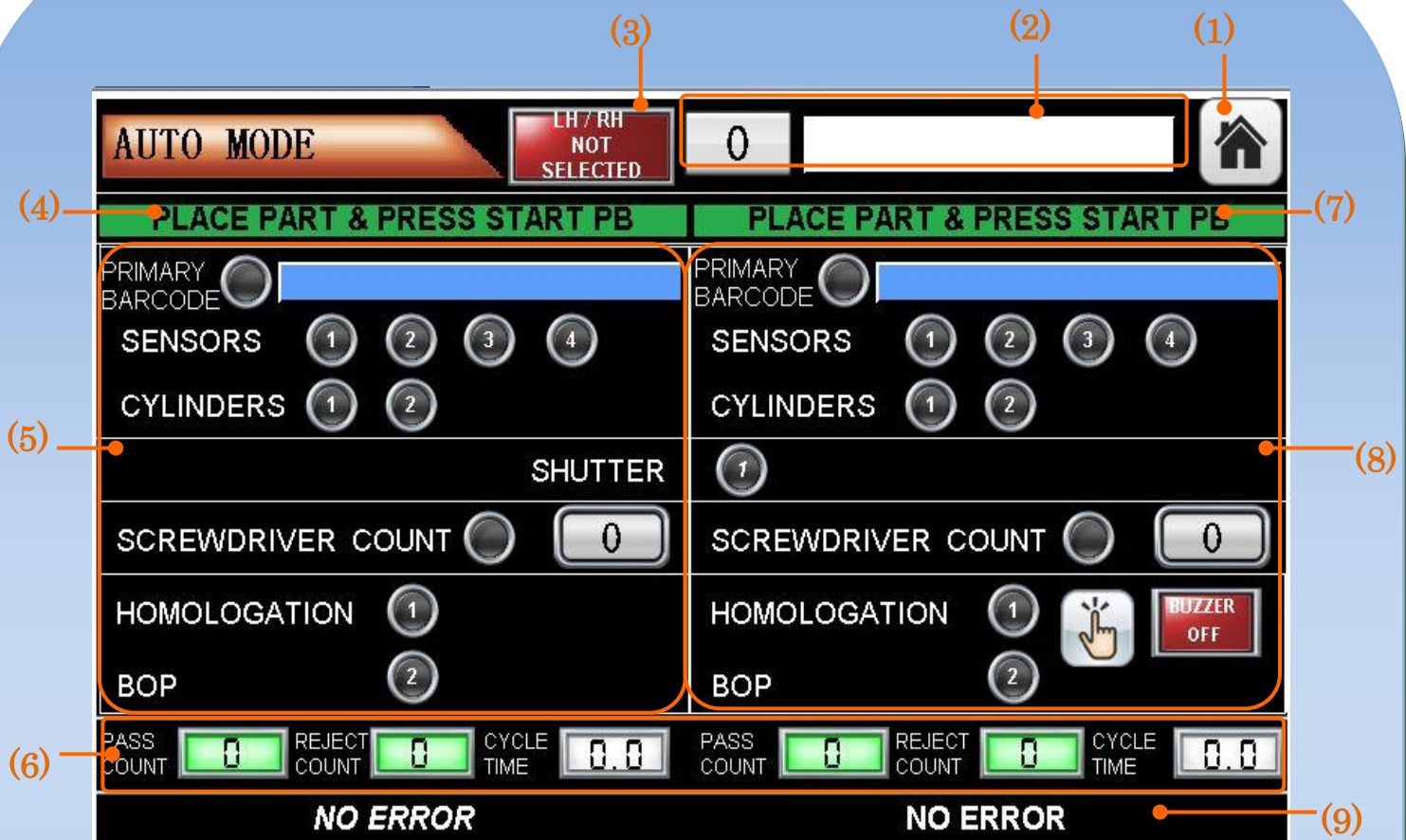
HMI SCREEN DETAILS:

HOME SCREEN



1. Program name and number display.
2. Program monitoring screen button .
3. Configuration setting screen navigation button.
4. Auto mode screen navigation button.
5. Date and time display.
6. Input/Output screen navigation button.
7. Manual mode screen navigation button.

AUTO MODE SCREEN



- 1.Home screen navigation button.
2. Program name & number display.
3. LH/RH not selected button.
4. LH Status message display.
5. LH primary barcode status display.
 - LH sensors status display.
 - LH cylinders status display.
 - Shutter cylinder status display.

- Screwdriver count and status display.
- Homologation & BOP status display.

6.LH/RH pass count, Reject count, cycle time operation status display.

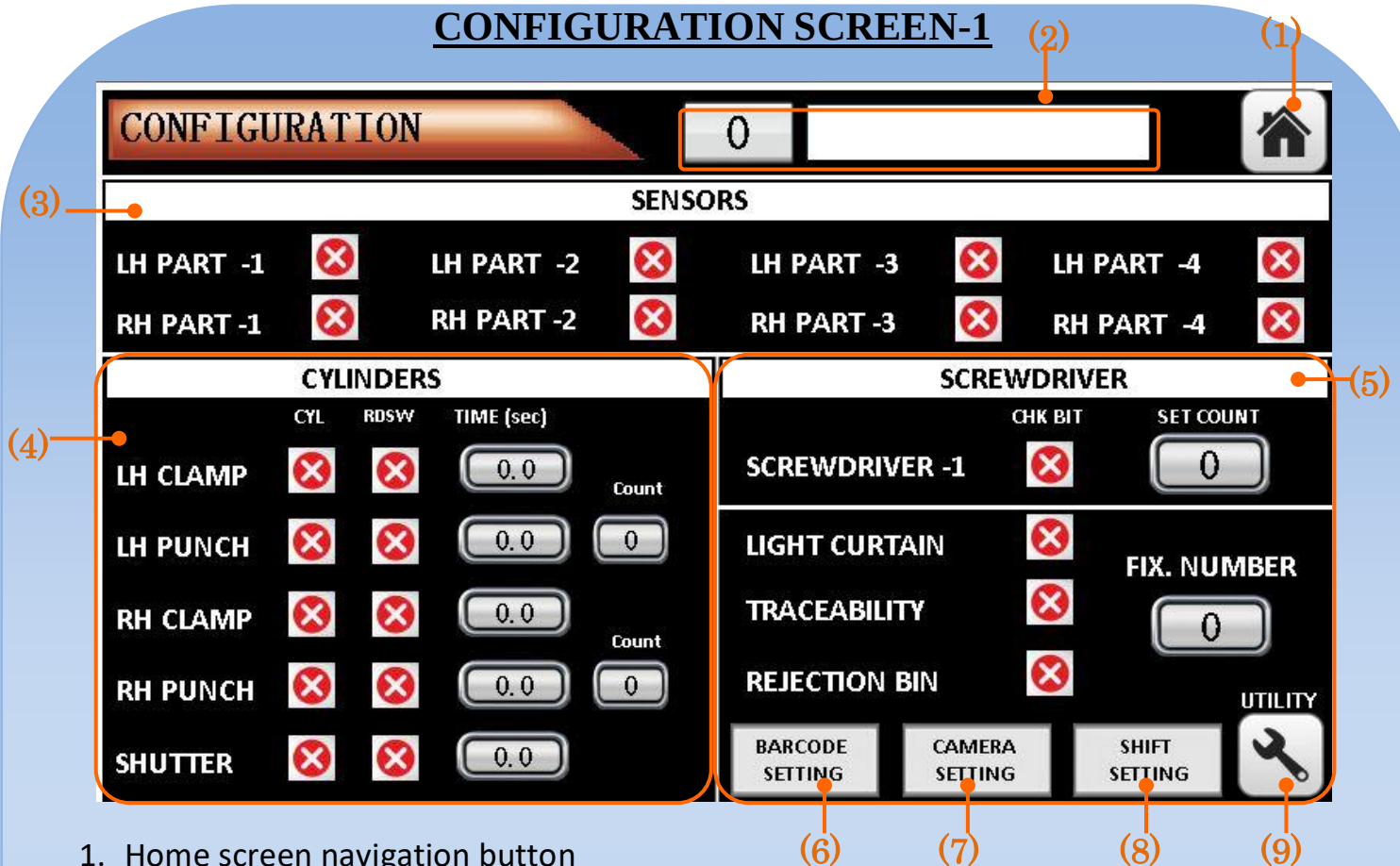
7. RH status message display.

8. RH primary barcode status display.

- LH sensors status display.
- LH cylinders status display.
- Shutter cylinder status display.
- Screwdriver count and status display.
- Homologation & BOP status display.

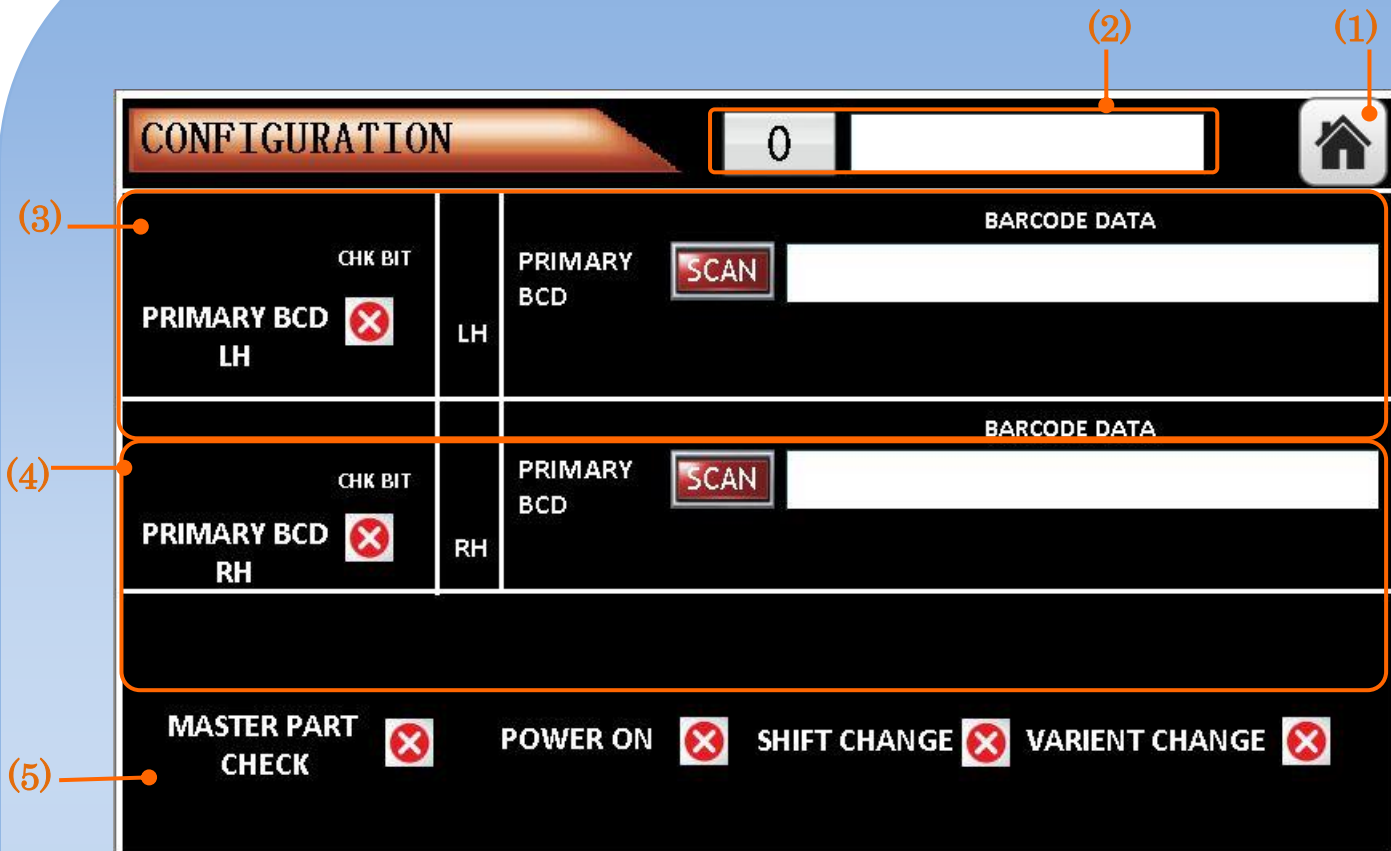
9.Error message display.

CONFIGURATION SCREEN-1



1. Home screen navigation button
2. Program name & number display.
3. LH/RH part present sensor select/unselect option.
4. LH/RH clamp, punch, shutter cylinder reed switch select/unselect option and time per second display.
5. Screwdriver select/unselect option and set count display.
 - Light curtain, Traceability and rejection bin select/unselect option and fixture number display.
6. Barcode setting navigation button.
7. Camera setting navigation button.
8. Shift setting navigation button.
9. Utility-Password and Date time setting.

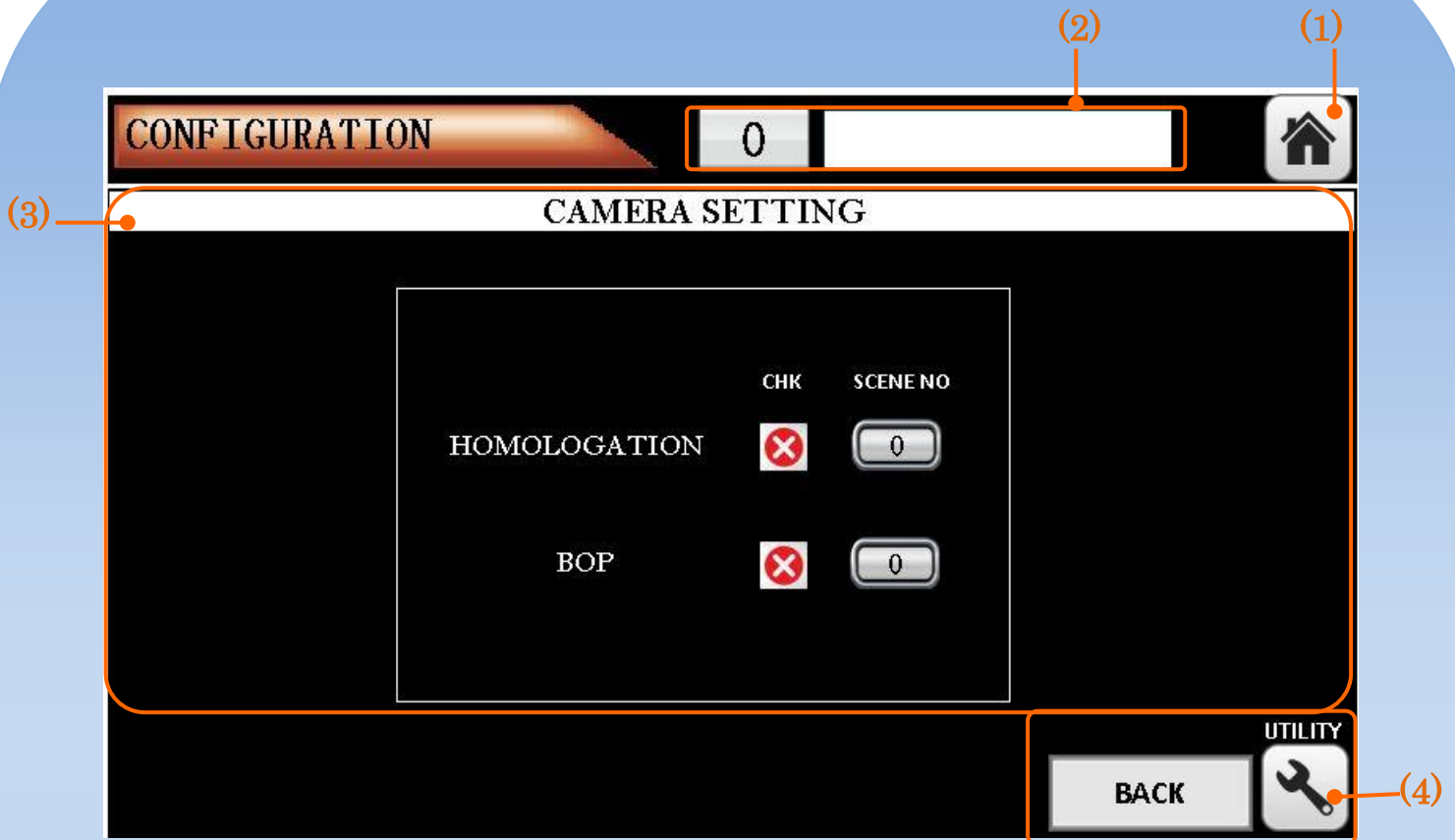
CONFIGURATION SCREEN-2



The screenshot shows the CONFIGURATION SCREEN-2 interface. At the top, there is a header bar with the title "CONFIGURATION" on the left, a display showing "0" in the center, and a home icon on the right. Below the header, the screen is divided into two main sections. The top section is for the LH (Left Hand) side, and the bottom section is for the RH (Right Hand) side. Each section contains a "PRIMARY BCD" label, a "CHK BIT" checkbox (both marked with a red 'X'), a "SCAN" button, and a "BARCODE DATA" input field. At the bottom of the screen, there is a row of four checkboxes: "MASTER PART CHECK", "POWER ON", "SHIFT CHANGE", and "VARIANT CHANGE", all of which are marked with a red 'X'. Numbered callouts (1) through (5) point to the following elements: (1) Home icon, (2) Program name & number display, (3) Primary BCD LH select/unselect option, (4) Primary BCD RH select/unselect option, and (5) Master part check, power on, shift change and variant change select/unselect option.

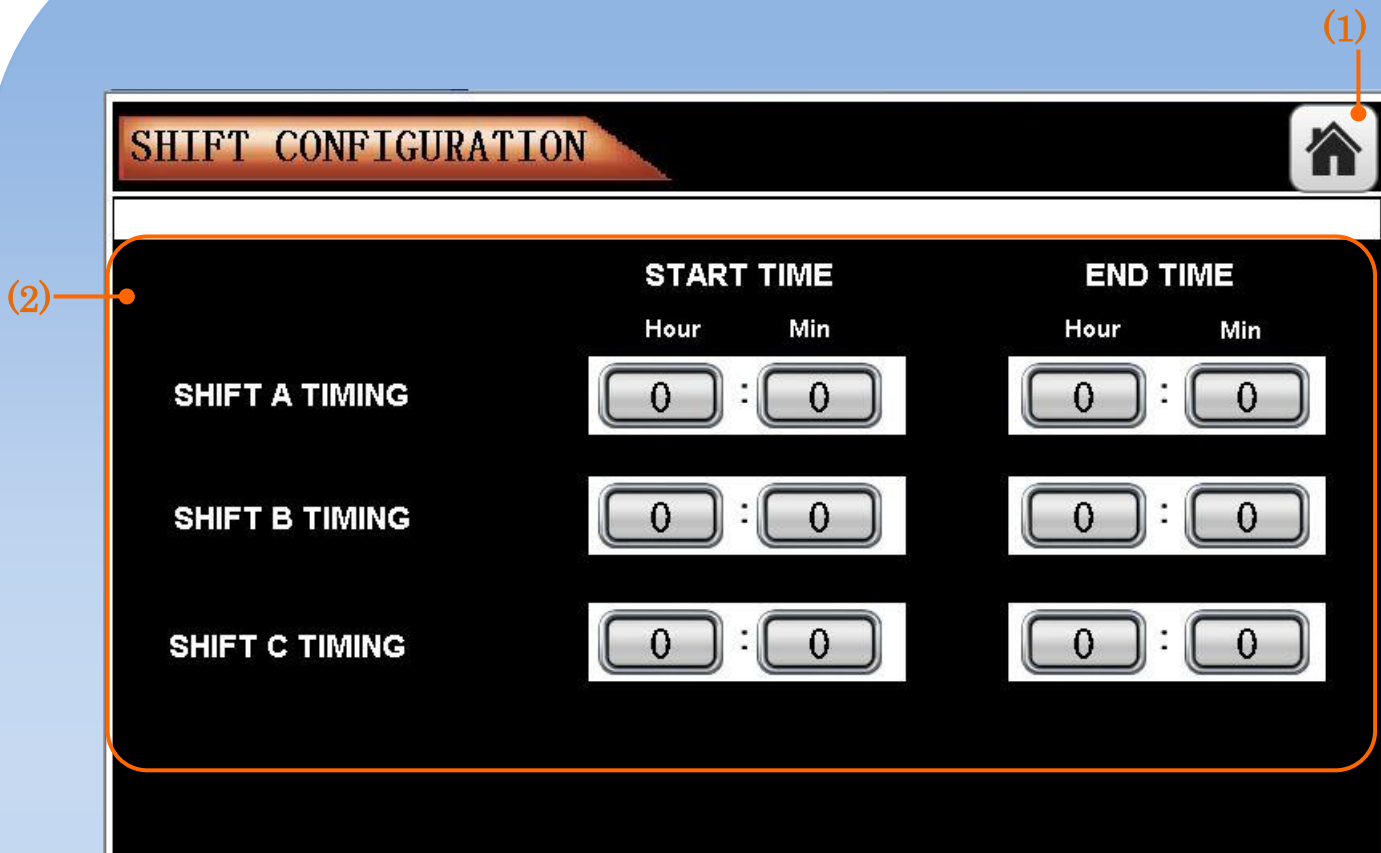
1. Home screen navigation button
2. Program name & number display.
3. Primary BCD LH select/unselect option.
 - Primary BCD scan navigation button and barcode data display.
4. Primary BCD RH select/unselect option.
 - Primary BCD scan navigation button and barcode data display.
5. Master part check, power on, shift change and variant change select/unselect option.

CONFIGURATION SCREEN-3



1. Home screen navigation button
2. Program name & number display.
3. Camera setting:-
 - Homologation select/unselect option and scene number display.
 - BOP select/unselect option and scene number display.
4. Back page navigation button.
 - Utility:-Password and date time setting.

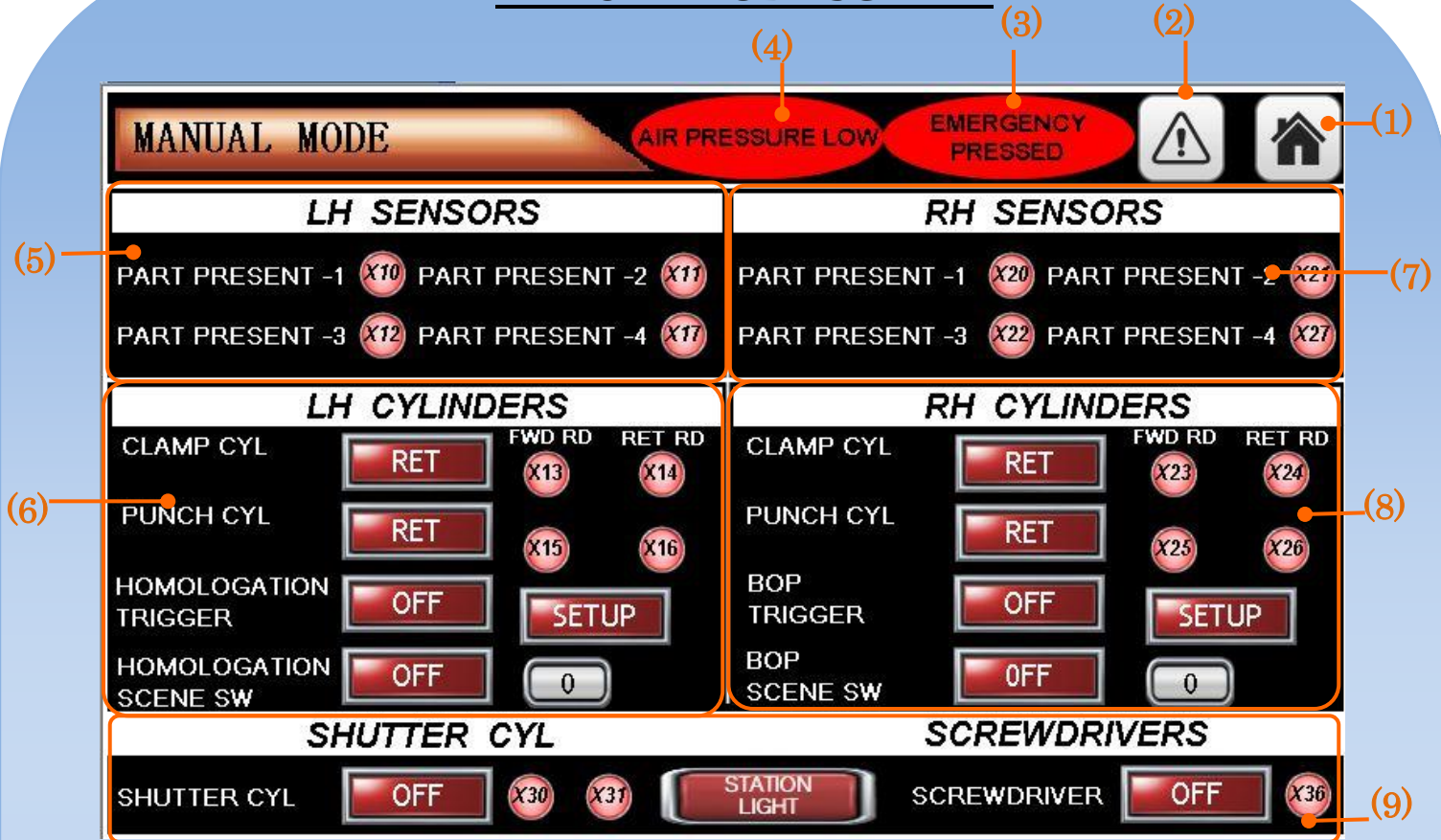
SHIFT CONFIGURATION SCREEN-4



	START TIME		END TIME	
	Hour	Min	Hour	Min
SHIFT A TIMING	0	0	0	0
SHIFT B TIMING	0	0	0	0
SHIFT C TIMING	0	0	0	0

1. Home screen navigation button
2. Shift A timing: -Start time and end time setting.
 - Shift B timing: -Start time and end time setting.
 - Shift C timing: -Start time and end time setting.

MANUAL MODE SCREEN



1. Home screen navigation button.
2. Error navigation option.
3. Emergency pressed status display.
4. Air pressure low status display.
5. LH sensor manual operation status display.
6. LH clamp and punch cylinders forward/return manual operation status display.
 - Homologation trigger on/off manual operation status display.
 - Homologation scene switch on/off manual operation status display.
7. RH sensors manual operation status display.



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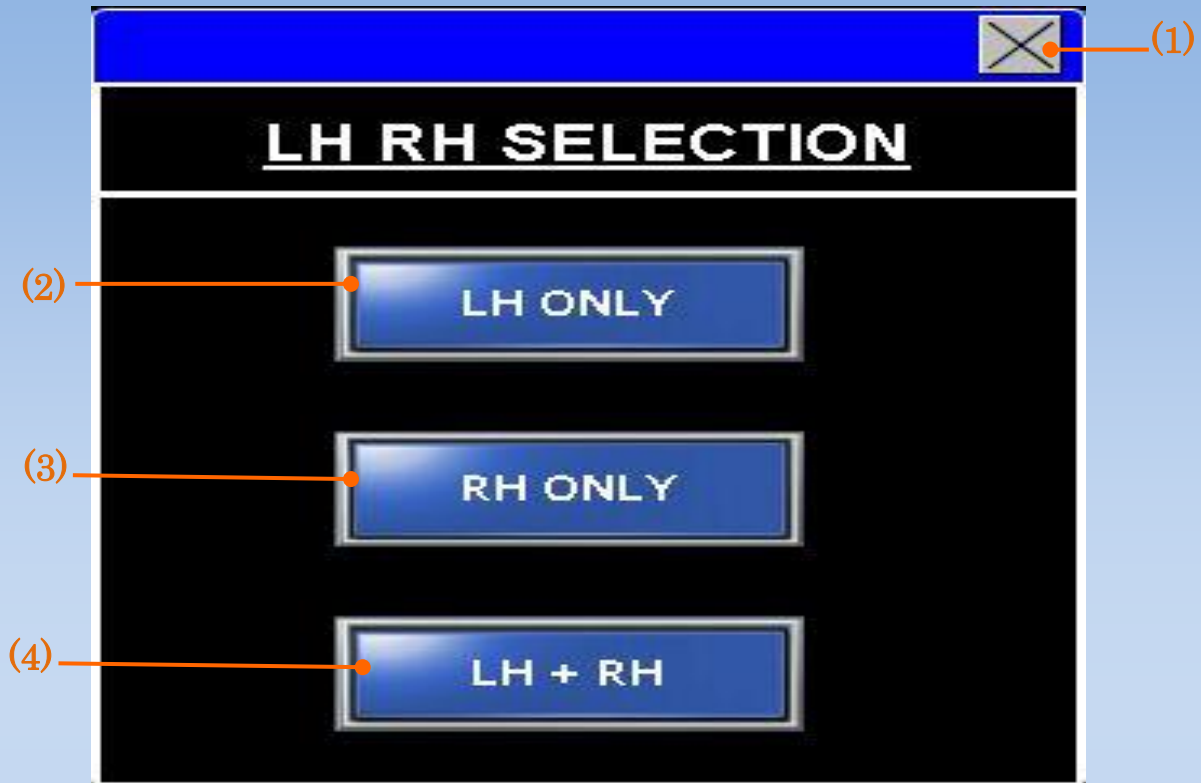
8. RH clamp and punch cylinders forward/return manual operation status display.

- Homologation trigger on/off manual operation status display.
- Homologation scene switch on/off manual operation status display.

9. Shutter cylinder on/off operation status display.

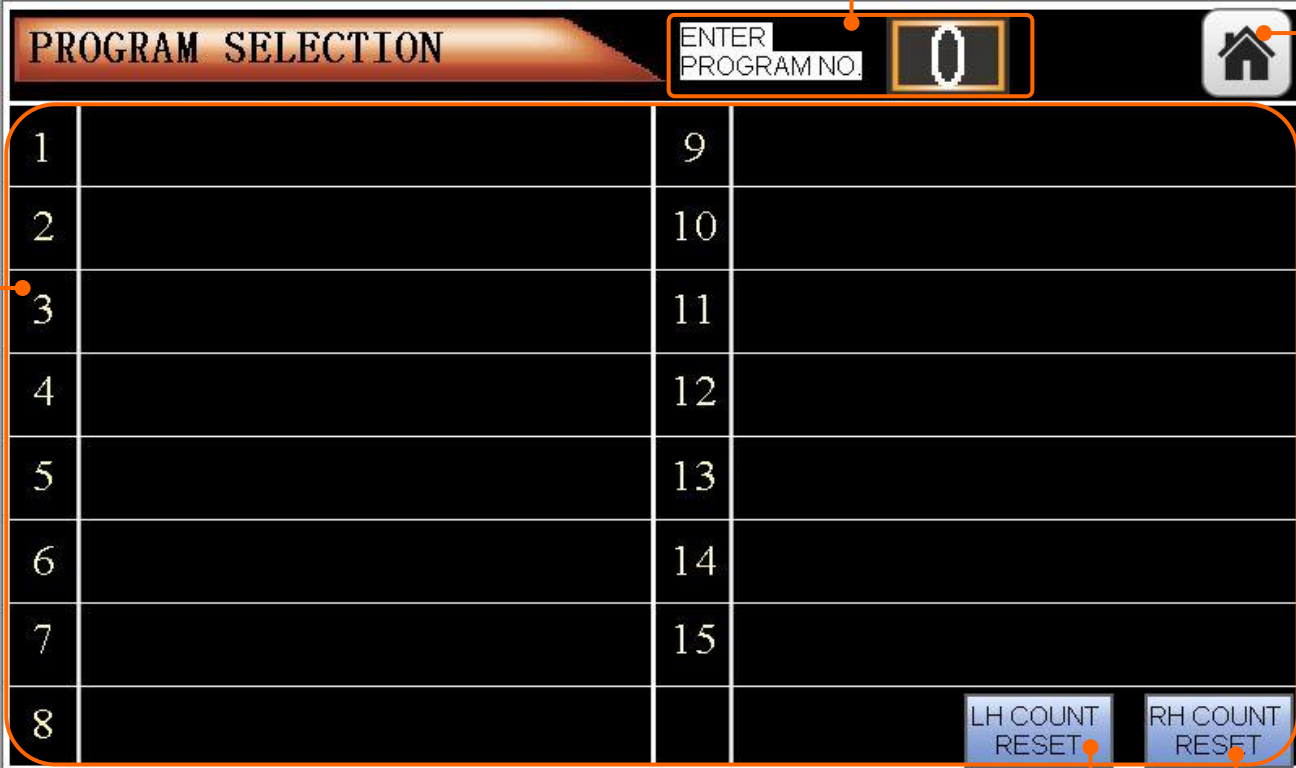
- Station light navigation button.
- Screwdriver on/off operation status display.

PROCESS SELECTION SCREEN



1. Close screen indication.
2. LH only select option.
3. RH only select option.
4. LH and RH both select option.

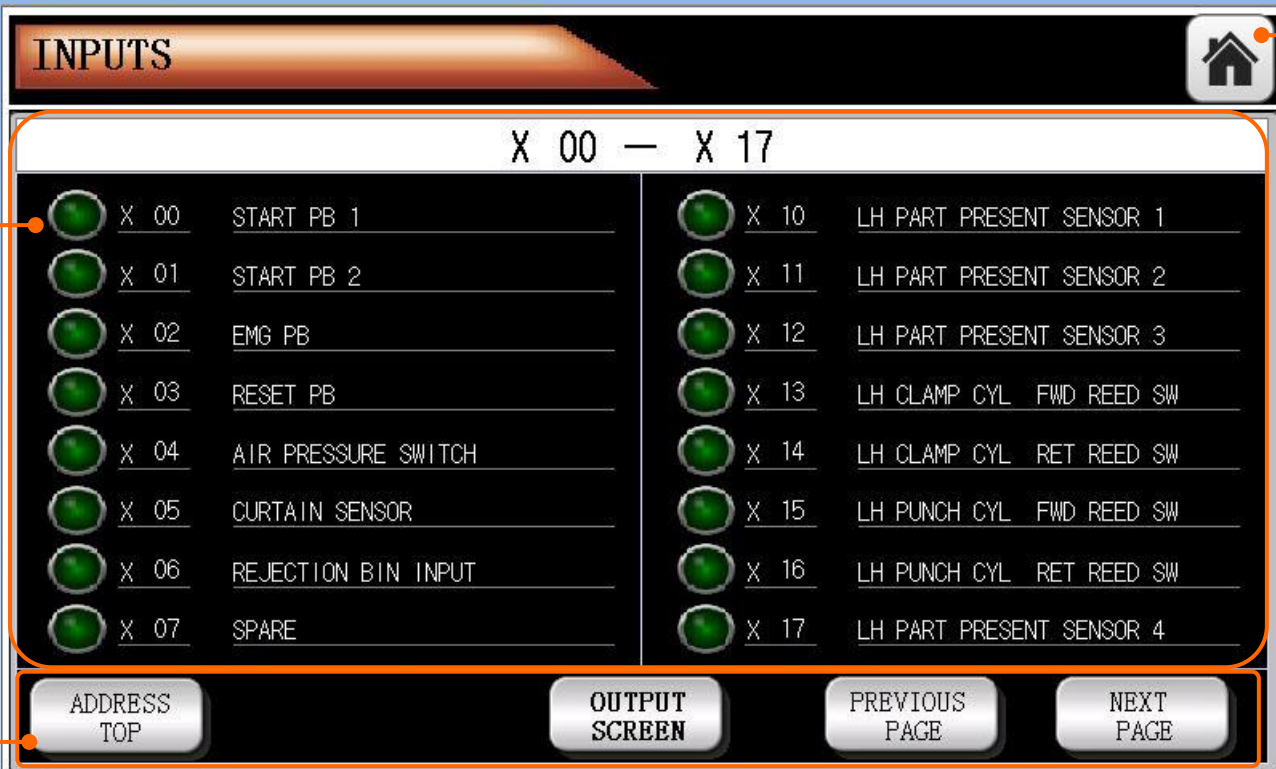
PROGRAM SELECTION SCREEN



The screenshot shows the 'PROGRAM SELECTION' screen. At the top left is the title 'PROGRAM SELECTION'. To its right is a label 'ENTER PROGRAM NO.' followed by a digital display showing the number '0'. On the far right is a home icon. Below these is a list of program numbers from 1 to 15, arranged in two columns. At the bottom right are two buttons labeled 'LH COUNT RESET' and 'RH COUNT RESET'. Numbered callouts point to: (1) Home icon, (2) 'ENTER PROGRAM NO.' label, (3) Program list area, (4) 'LH COUNT RESET' button, and (5) 'RH COUNT RESET' button.

1. Home screen navigation button.
2. Enter program number and display.
3. Program name and number list.
4. LH count reset navigation button.
5. RH count reset navigation button.

INPUT STATUS-1



INPUTS

X 00 — X 17

X 00	START PB 1	X 10	LH PART PRESENT SENSOR 1
X 01	START PB 2	X 11	LH PART PRESENT SENSOR 2
X 02	EMG PB	X 12	LH PART PRESENT SENSOR 3
X 03	RESET PB	X 13	LH CLAMP CYL FWD REED SW
X 04	AIR PRESSURE SWITCH	X 14	LH CLAMP CYL RET REED SW
X 05	CURTAIN SENSOR	X 15	LH PUNCH CYL FWD REED SW
X 06	REJECTION BIN INPUT	X 16	LH PUNCH CYL RET REED SW
X 07	SPARE	X 17	LH PART PRESENT SENSOR 4

ADDRESS TOP OUTPUT SCREEN PREVIOUS PAGE NEXT PAGE

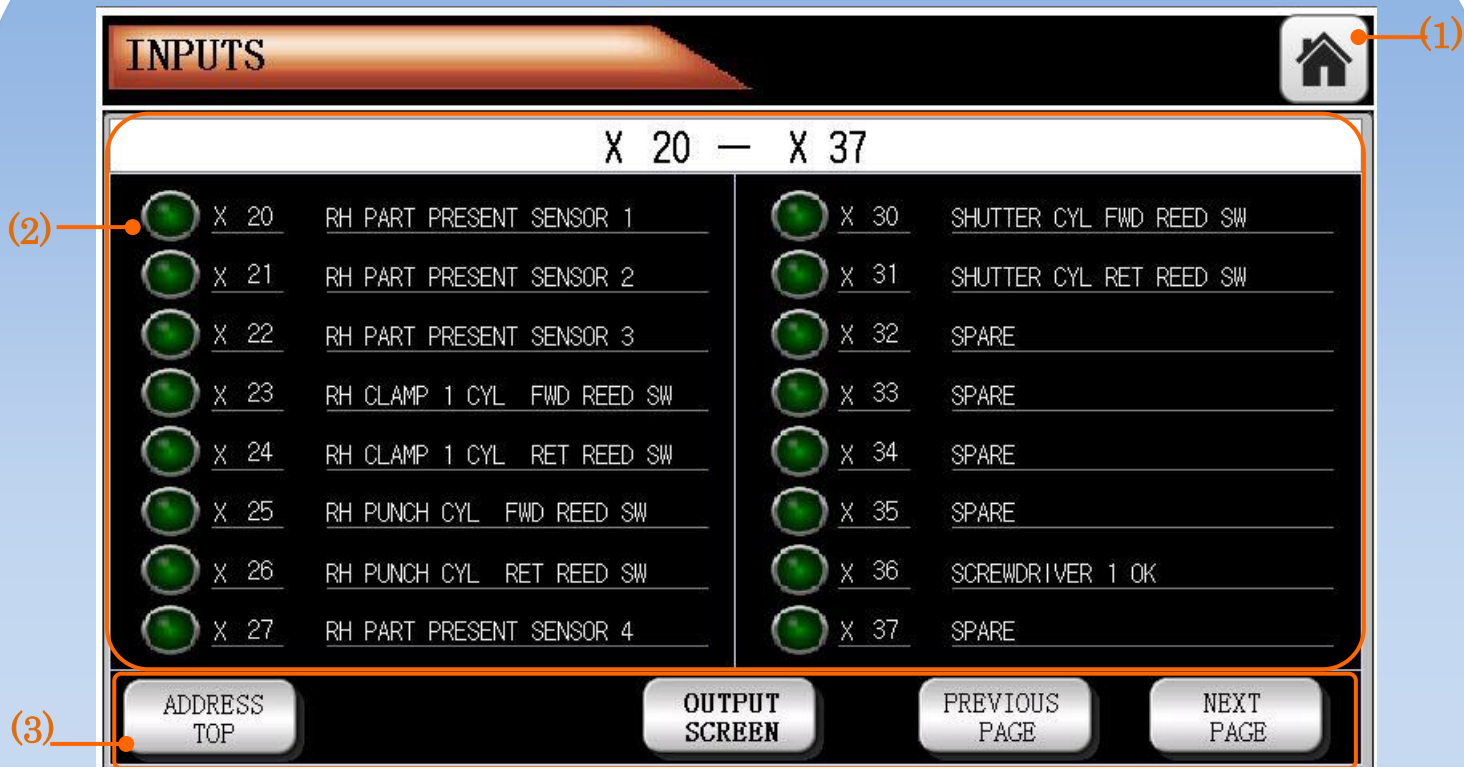
1. Home screen navigation button.

2. Plc input status.

Add press top navigation button.

- Output screen navigation button.
- Previous page navigation button.
- Next page navigation button.

INPUT STATUS-2



INPUTS

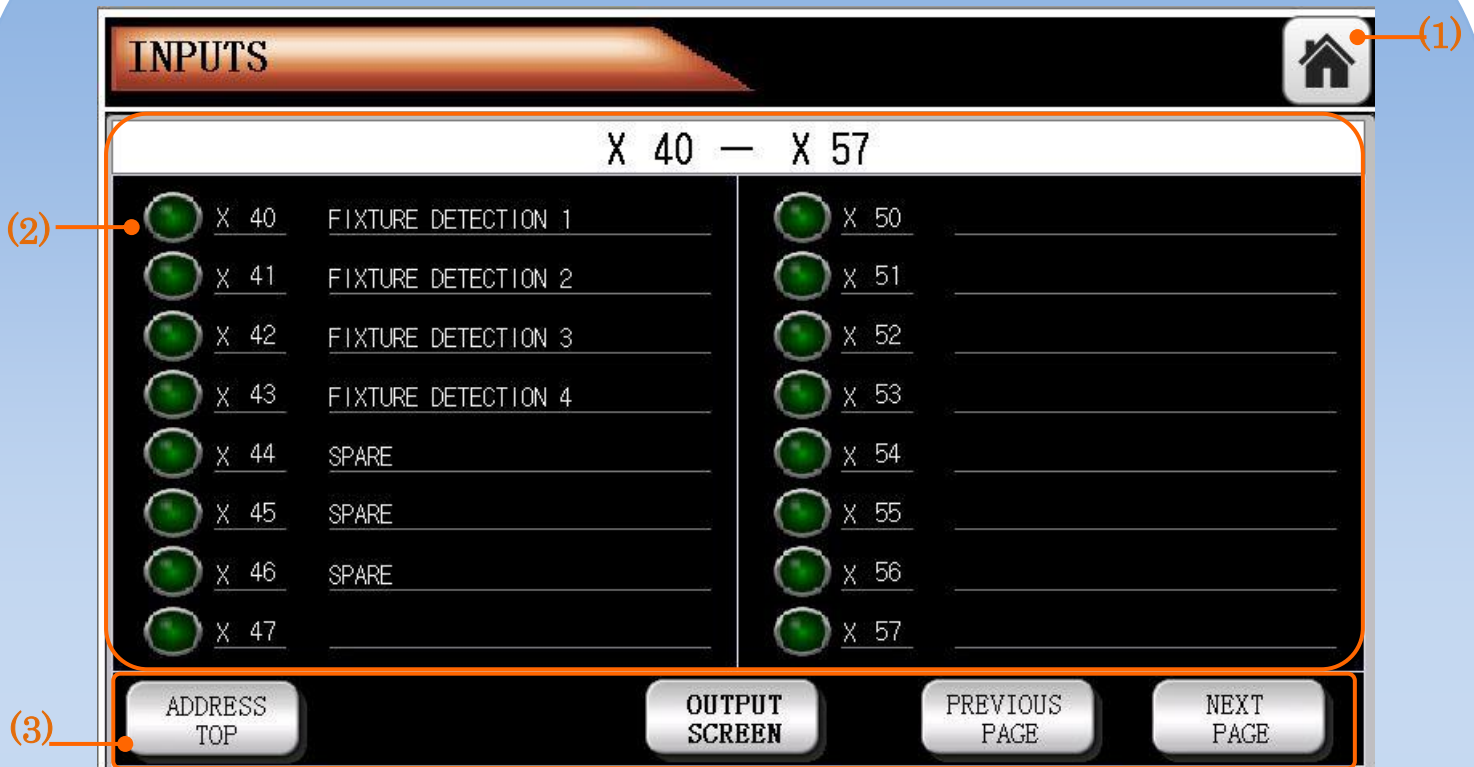
X 20 — X 37

X 20	RH PART PRESENT SENSOR 1	X 30	SHUTTER CYL FWD REED SW
X 21	RH PART PRESENT SENSOR 2	X 31	SHUTTER CYL RET REED SW
X 22	RH PART PRESENT SENSOR 3	X 32	SPARE
X 23	RH CLAMP 1 CYL FWD REED SW	X 33	SPARE
X 24	RH CLAMP 1 CYL RET REED SW	X 34	SPARE
X 25	RH PUNCH CYL FWD REED SW	X 35	SPARE
X 26	RH PUNCH CYL RET REED SW	X 36	SCREWDRIVER 1 OK
X 27	RH PART PRESENT SENSOR 4	X 37	SPARE

ADDRESS TOP OUTPUT SCREEN PREVIOUS PAGE NEXT PAGE

1. Home screen navigation button.
 2. Plc input status.
 3. Add press top navigation button.
- Output screen navigation button.
 - Previous page navigation button.
 - Next page navigation button.

INPUT STATUS-3



1. Home screen navigation button.
2. Plc input status.
3. Add press top navigation button.

- Output screen navigation button.
- Previous page navigation button.
- Next page navigation button.

OUTPUT STATUS-1



OUTPUTS

Y 00 — Y 17

☒ Y 00	PASS LAMP	☒ Y 10	LH CLAMP CYL RET SOL
☒ Y 01	PROCESS LAMP	☒ Y 11	LH PUNCH CYL FWD/RET SOL
☒ Y 02	ERROR LAMP	☒ Y 12	SPARE
☒ Y 03	BUZZER	☒ Y 13	RH CLAMP CYL FWD SOL
☒ Y 04	SPARE	☒ Y 14	RH CLAMP CYL RET SOL
☒ Y 05	AIR OFF SOL	☒ Y 15	RH PUNCH CYL FWD/RET SOL
☒ Y 06	STATION LIGHT	☒ Y 16	SPARE
☒ Y 07	LH CLAMP CYL FWD SOL	☒ Y 17	SHUTTER CYL FWD SOL

ADDRESS TOP INPUT SCREEN PREVIOUS PAGE NEXT PAGE

1. Home screen navigation button.
 2. Plc output status.
 3. Address top navigation button.
- Input screen navigation button.
 - Previous page navigation button.
 - Next page navigation button.

OUTPUT STATUS-2



Y 20 — Y 37	
Y 20	SHUTTER CYL REV SOL
Y 21	SCREWDRIIVER ON
Y 22	SPARE
Y 23	SPARE
Y 24	SPARE
Y 25	SPARE
Y 26	SPARE
Y 27	SPARE
Y 30	SPARE
Y 31	SPARE
Y 32	SPARE
Y 33	SPARE
Y 34	SPARE
Y 35	SPARE
Y 36	SPARE
Y 37	SPARE

Navigation buttons: ADDRESS TOP, INPUT SCREEN, PREVIOUS PAGE, NEXT PAGE

1. Home screen navigation button.
 2. Plc output status.
 3. Address top navigation button.
- Input screen navigation button.
 - Previous page navigation button.
 - Next page navigation button.

OUTPUT STATUS-3



Y 40 — Y 57	
Y 40 SPARE	Y 50 SPARE
Y 41 SPARE	Y 51 SPARE
Y 42 SPARE	Y 52 SPARE
Y 43 SPARE	Y 53 SPARE
Y 44 SPARE	Y 54 SPARE
Y 45 SPARE	Y 55 SPARE
Y 46 SPARE	Y 56 SPARE
Y 47 SPARE	Y 57 SPARE

Navigation buttons: ADDRESS TOP (3), INPUT SCREEN, PREVIOUS PAGE, NEXT PAGE

1. Home screen navigation button.
 2. Plc output status.
 3. Address top navigation button.
- Input screen navigation button.
 - Previous page navigation button.
 - Next page navigation button.

ALARM HISTORY



1. Home screen navigation button.

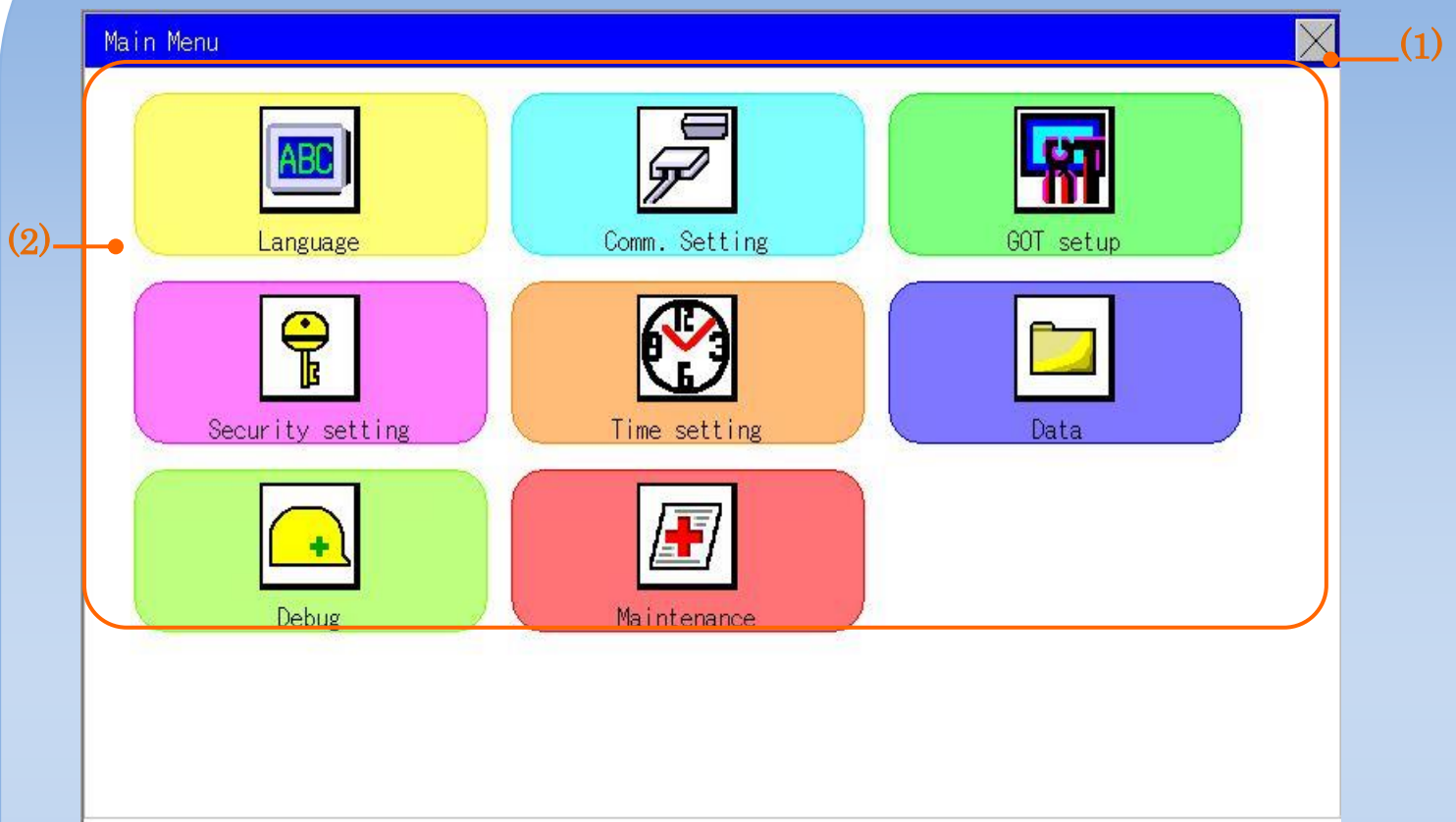
2. Alarm history list.

- Page up and row up navigation option.

3. Delete navigation button.

- Delete all navigation button.
- Check navigation button.
- Check all navigation button.
- Circuit navigation button.

UTILITY SETTING



2. Close screen indication.
3. Language navigation option.
 - Communication setting navigation option.
 - Got setup setting navigation option
 - Security setting navigation option.
 - Time setting navigation option.
 - Data setting navigation option.
 - Debug setting navigation option.
 - Maintenance navigation button.



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MACHINE ERRORS

SR.N O	ERROR DESCRIPTION		CAUSES OF ERROR		REMEDY
1	HOME ERROR!!!	1	There may be any pneumatic cylinder not in home position.	1	Set all retract reed sensor in correct position.
		2	The sensor position might have shifted.	2	Set sensor properly.
		3	There may be loose connection in the sensor connector.	2	Tighten the sensor connector and press reset push button to clear the error.
2	EMERGENCY PB PRESSED	1	Emergency push button might have pressed.	1	Release emergency push button and press Reset push button to clear the error.
		2	There may be loose connection in the push button contact.	2	Tighten the electrical connection and press Reset push button to clear the error.
3	LOW AIR PRESSURE	1	Check the pneumatic supply.	1	Adjust the pressure to optimum level.
		2	There may be loose connection in the pressure indicator.	2	Tighten the connector and press Reset push button to clear the error.
4	FIXTURE ERROR	1	Check loaded fixture and set fixture in hmi are same or different.	1	Go in configuration screen then click on fixture number button to set actual fixture number.
		2	Check input indication on plc there may be loose connection in the electrical connector.	2	Tighten the electrical connection and press Reset push button to clear the error.
5	PRIMARY BARCODE ERROR	1	The scan data is not mach with actual set data in configuration screen	1	Check scan barcode data is ok or not
		2	There may be barcode are not scan properly.	2	Check the proper condition of the barcode and scan properly.
		3	There may be primary barcode are not scanned.	4	Scan primary barcode properly.
6	BARCODE NOK	1	The scan data is not mach with actual set data in configuration screen	1	Check scan barcode data is ok or not
		2	The scanner may not be able to find the right data after scanning.	2	Check connection of barcode scanner or barcode scan properly.



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		3	There may be loose communication cable connection.	3	Check connection of scanner are connect correct or wrong.
		4	There may be barcode are not scan properly.	4	Check the proper condition of the barcode and scan properly.
7	PART SENSOR 1 ABSENT	1	Check sensing part is present or absent	1	If sensing part is present please check sensor is properly teach on sensing part
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
8	PART SENSOR 2 ABSENT	1	Check sensing part is present or absent	1	If sensing part is present please check sensor is properly teach on sensing part
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
9	PART SENSOR 3 ABSENT	1	Check sensing part is present or absent	1	If sensing part is present please check sensor is properly teach on sensing part
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
10	CLAMP CYL FWD ERROR	1	The reed sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
11	CLAMP CYL RET ERROR	1	The reed sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
12	SHUTTER CYL FWD ERROR	1	The reed sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error



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		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
12	SHUTTER CYL RET ERROR	1	The sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
12	DOT PUNCH CYL FWD ERROR	1	The reed sensor position might have shifted.	1	Position the sensor properly.
		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
		3	The cylinder might not have reached its position fully.	3	Adjust the pressure to optimum level.
13	LIGHT CURTAIN DISTURB	1	There may be object or hand in between safety light curtain.	1	Remove the object or hand from the safety light curtain and press reset PB for reset error.
14	HOMOLOGATION COMMUNICATION ERROR	1	There may be loose connection in communication cable.	1	Check communication cable between camera controller and plc.
15	BOP COMMUNICATION ERROR	1	There may be loose connection in communication cable.	1	Check communication cable between camera controller and plc.
16	HOMOLOGATION NOK	1	The part position might have shifted.	1	Position the part properly.
		2	Dirt may have accumulated on part.	2	Clean part with cotton.
17	BOP NOK	1	The part position might have shifted.	1	Position the part properly.
		2	check the bop part is present	2	if bop part is absent place part and check again
18	PLACE PART IN REJECTION BIN	1	There may be NOK part not put in rejection been.	1	Put NOK part in rejection been and press reset push button for reset the error.
19	PART SENSOR 4 ABSENT	1	The sensor position might have shifted.	1	Position the sensor properly.



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		2	There may be loose connection in the sensor connector	2	Tighten the sensor connector and press reset push button to clear the error
20	SERVER NOT CONNECTED	1	There may be loose connection in the server communication cable. Or server not connected	1	Check server communication cable connected properly or not. Or reset server using server reset button on auto screen.
21	SERVER BUSY	1	Data not send or receive properly.	1	Check all communication device are ON condition or check connection of all communication cable.
22	PLACE NEW PART	1	After cycle complete the previous part are present still there.	1	Remove old part and place new part properly.
23	Log unknown error station not valid	1	There might be possibility of any unknown communication happen with server.	1	Check log window, contact supplier.
24	Previous Station NOK	1	There may be previous station cycle are not ok.	1	Run back to the previous station with all ok condition.



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DO'S & DONT'S:

Every effort has been taken to ensure user safety while designing this machine. However, you are requested to adhere to following instructions to avoid any mishap.

- The machine should not be used for any other purpose other than that specified in the intended use.
- Do not operate the machine without understanding the installation, operation and maintenance manual thoroughly.
- Only trained, skilled, qualified and authorized personnel should install, operate and maintain the machine.
- Do not commission the machine without the aid of the qualified instructor.
- Do not reach into moving machinery.
- Never perform complex maintenance procedure single handed. Especially during start-ups as it is necessary to have another person manning the emergency stop in case of any problem.
- No tools other than that are prescribed should be used for machine maintenance.
- Do not store any loose tools in the machine compartments or control panel.
- If the machine is down for maintenance, place a sign over the control panel to inform others about that.
- Switch off the machine while connecting/disconnecting into electrical connector.



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